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Intervening against the Fed[☆]

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A R T I C L E I N F O

Dataset link: Replication Package for "Intervening against the Fed" (Original data)

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A B S T R A C T

This paper studies the effectiveness and mechanism of foreign exchange interventions (FXIs) for mitigating US monetary policy spillovers. Without interventions, contractionary US monetary policy shocks trigger foreign exchange depreciations, raise risk premiums, and induce portfolio outflows, thereby reducing foreign stock prices. The stock prices of firms with US dollar debt decline significantly more, indicating a strong role for a balance sheet channel. However, "intervening against the Fed" stabilizes exchange rate depreciations, the risk premium, and portfolio flows. FXIs also disproportionately cushion the reduction in stock prices for firms with US dollar debt, muting the balance sheet channel of exchange rates.

1. Introduction

The Federal Reserve is an important driver of the "Global Financial Cycle" and changes in US monetary policy can have major spillovers to the global economy. When the Fed tightens monetary policy, the US dollar appreciates, and global stock prices as well as credit decline (Rey, 2015; Miranda-Agrippino and Rey, 2020; Gürkaynak et al., 2021). Although both academics and policymakers have signaled the first-order importance of how countries could protect themselves against this unexpected tightening of monetary policy, there is limited consensus on the effectiveness of different measures.

Foreign exchange interventions (FXIs) are an increasingly popular policy tool among central banks attempting to insulate themselves from those spillovers. For instance, on September 21, 2022, the Federal Open Market Committee (FOMC) announced to increase the Fed funds rate by 0.75 percentage points, after which the Japanese yen depreciated strongly. However, when the Bank of Japan intervened in the FX market by selling the US dollar and buying the Japanese yen, the depreciation immediately reversed (Fig. 1). This example shows how "intervening against the Fed" can potentially offset US monetary spillovers to exchange rates. Despite their widespread use, systematic evidence of how FXIs function as a tool to insulate financial markets and the economy from US monetary policy spillovers is elusive. We aim to fill this gap in the literature.

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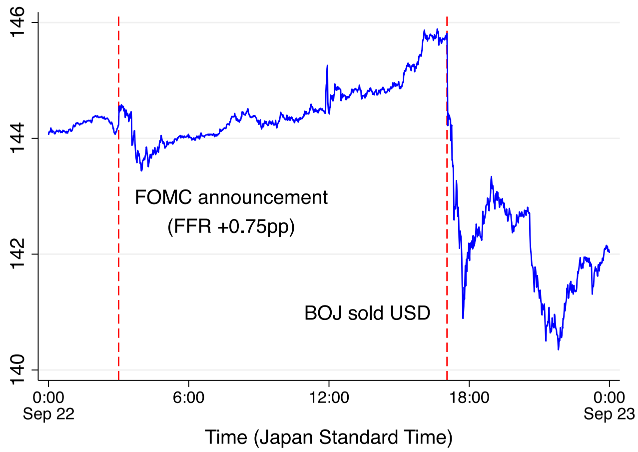

Fig. 1. Spot exchange rate: US Dollar to Japanese Yen.
Note: The figure reports the minute-by-minute US Dollar to Japanese yen spot exchange rate on September 22, 2022. The exchange rate is defined as the value of one US Dollar in terms of yen, and a higher value implies the appreciation of the Dollar or depreciation of Japanese yen.
Source: Datastream.

We first provide evidence on how US monetary policy spills over if countries do not counter-intervene. When US monetary policy unexpectedly tightens, domestic exchange rates depreciate against the US dollar, the currency risk premium rises, and investors withdraw their portfolios from foreign countries. Furthermore, US monetary policy tightening leads to a fall in stock prices, which is disproportionately so for firms that borrow in US dollars. The depreciation of the domestic exchange rate reduces firms' net worth due to higher debt repayment, leading to lower stock prices (Krugman, 1999; Céspedes et al., 2004). Then we show that intervening against the Fed, by unexpectedly selling the US dollar in response to a contractionary monetary policy shock, mitigates exchange rate depreciation and stock price declines, but more so for firms with US dollar debt. When US monetary policy tightens and central banks counter-intervene, exchange rates and stock prices for firms with and without US dollar debt remain statistically and economically unchanged. These results suggest that intervening against the Fed mutes the balance sheet channel of exchange rates triggered by US monetary policy and can protect countries from exposure to the Global Financial Cycle. Moreover, intervening against the Fed successfully prevents a rise in the currency risk premium and portfolio equity outflows, further stabilizing financial markets over and above the balance sheet channel.

Recent theory suggests that FXIs can affect the exchange rate and improve welfare (Gabaix and Maggiori, 2015; Cavallino, 2019; Amador et al., 2020; Fanelli and Straub, 2021; Maggiori, 2022; Itskhoki and Mukhin, 2023). However, empirical evidence on the effectiveness of FXIs is limited by identification challenges. Both US monetary policy and FXIs are endogenously conducted, taking current and future information into account. Moreover, changes in monetary or FX policy are inextricably linked and hard to isolate from other factors. These problems make it harder to identify the channels through which US monetary policy spills over to other countries and whether FXIs are successful in mitigating them. We overcome these issues in several ways.

Our novel identification method employs a multi-event study high-frequency approach by combining daily data on FXIs with firm-level stock returns, exchange rates, as well as risk premia and portfolio flows around US monetary policy decisions.[1] In particular, we use an event-study local projection difference-in-differences (LP-DID) approach in the spirit of Dube et al. (2025) around each FOMC meeting. The advantages of using financial market data around FOMC events are twofold. First, compared to outcomes only available at lower frequencies, such as investment, they can be measured at a daily frequency, which mitigates the aggregation bias introduced by aggregating monetary policy shocks or FXIs to lower frequencies. Second, market prices are forward-looking and therefore respond more quickly as they incorporate expected future economic outcomes, such as slower-moving variables like firms' investments and profits, that would not appear in contemporaneous balance sheet measures. Third, our identification strategy helps to identify a counterfactual of what would have happened if countries had not intervened. Simply observing exchange rates and stock prices around an intervention without comparing them to a counterfactual can lead to misleading conclusions.

To measure the effect of interventions, we use daily FXI data from 13 countries. We define expected and unexpected counter-interventions as situations in which the central bank sells (buys) US dollars when the Fed unexpectedly tightens (loosens) policy. We call this policy "intervening against the Fed" as selling US dollars in exchange for domestic currency is intended to put appreciation pressure on the domestic exchange rate by absorbing its supply and vice versa. We then define intervening against the Fed if the central bank counter-intervenes within five days after the FOMC meeting. To address the potential concern that FXIs may be driven by economic fundamentals that could bias our estimates, we also estimate deviations from an FX counter-intervention policy rule incorporating observable and unobservable macroeconomic and financial characteristics, as well as historical responses to Federal Reserve policy. Deviations from the FXI policy rule can be interpreted as unexpected FXIs, helping to isolate the effects of potential confounding factors and estimate the direct effects of interventions.

For US monetary policy, we use cleanly identified high-frequency shocks by Nakamura and Steinsson (2018). The high-frequency monetary shocks are estimated by a change in Fed funds futures in a 30-minute time window around the FOMC announcement so that monetary shocks are orthogonal to the limited amount of information revealed in this narrow time window. Using this monetary surprise can be seen as an external driver of changes in the exchange rate, allowing us to compare the responses of exchange rates and stock prices to FOMC announcements in countries that do and do not counter-intervene against the US dollar shortly after the announcement.

There are many channels through which the US monetary policy can spill over to other countries under integrated financial and goods markets. Higher US monetary policy rates, ceteris paribus, predict an appreciation of the US dollar on impact due to interest rate differentials in the uncovered interest parity (UIP) equation. In the financial market, this dollar appreciation lowers stock prices through a balance sheet channel by increasing debt repayments for firms that borrow in US dollars (Gopinath and Stein, 2021). Even if firms do not have dollar debt, monetary policy can affect stock prices by increasing currency risk premia (Lustig et al., 2011) or triggering portfolio flows (Converse and Mallucci, 2023; Converse et al., 2023). In the goods market, according to an expenditure switching channel, the depreciation of domestic exchange rates relative to the US dollar would then increase exports and raise the stock prices of firms (Mundell, 1957; Fleming, 1962). In addition, a tighter US monetary policy may lower demand and imports in the US, with adverse effects on foreign economies (Gourinchas, 2018).

[1] We refer to daily data as high-frequency. For the market microstructure literature, this definition may be inappropriate, but as the international finance literature predominantly focuses on monthly or quarterly changes, we consider our approach high-frequency within that tradition.

Our high-frequency approach, coupled with firm-level stock prices and information on the currency decomposition of corporate debt, allows us to compare the cross-sectional heterogeneity in firms' stock price responses within each country, at a given point in time, to a contractionary monetary policy shock, ruling out the possibility that other macroeconomic factors or other channels, such as the risk premium and portfolio rebalancing channel, are driving the response. We find that, when countries do not counter-intervene against the Fed, stock prices of firms with US dollar debt decline immediately after US monetary policy unexpectedly tightens. Quantitatively, a 10 basis point surprise increase in the federal funds rate is associated with a 3% decline in stock prices for firms with US dollar debt on the day of the FOMC meeting, building up to almost 10% after three days. In contrast, firms without US dollar debt see a decline in the stock price of less than 1%. This is explained by an increase in the US equity and currency risk premiums, as well as equity portfolio outflows, which affect both firms with and without dollar debt. The stark differential response, however, strongly suggests that US monetary policy affects stock prices abroad through a balance sheet channel.

To shed more light on this channel, we move toward the effect of US monetary policy on exchange rates. If the balance sheet channel of US monetary policy is at work, one would expect the behavior of exchange rates to mirror that of the stock price for firms with US dollar debt. And, indeed, for countries that do not counter-intervene, exchange rates depreciate strongly after the US monetary policy shock. Quantitatively, a 10 basis point increase in the US monetary policy rate leads to a persistent 2%–3% depreciation of foreign currency. We contribute to the literature by providing supporting evidence in favor of the effectiveness of FXIs on both the stock market and exchange rates, making progress on the endogeneity issues discussed by Gabaix and Maggiori (2015) and Maggiori (2022).

The depreciation of the exchange rate may also trigger an expenditure switching channel, increasing demand from foreigners due to lower prices, in which case exporting firms would benefit disproportionately (Mundell, 1957; Fleming, 1962). However, the contractionary US monetary policy shock reduces demand from the US via intertemporal substitution and therefore also demand for exports, potentially offsetting the positive effect from the exchange rate (Gourinchas, 2018).[2] When countries do not intervene and the exchange rate depreciates, the stock prices of exporters remain stable while those of non-exporters fall, suggesting that the negative demand effects from the contractionary monetary policy shock and the positive effects from the expenditure switching channel offset each other.

Turning to the role of FXIs in US monetary policy spillover, we estimate the effect of "intervening against the Fed" on the stock prices across the firm distribution, as well as the exchange rate, currency risk premiums, and portfolio flows. By tracking the effect of the FXIs within a short window after the FOMC meeting, we likely capture the central bank's decision to "intervene against the Fed". If, instead, we studied the unconditional effects of FXIs on exchange rates, it would be more difficult to understand the causal effect of the interventions. For instance, it may be possible that the central bank intervenes because it receives a signal about the economy that is unobservable, making it more difficult to understand the counterfactual response of the exchange rate if the central bank had not intervened. As we employ an event study in a short window around the FOMC announcement, the market movements are likely driven by US monetary policy and, without the FOMC decision, would have remained systematically unchanged, as the pre-trends indicate, allowing us to estimate a more precise counterfactual of not intervening.

Using these unexpected monetary shocks and FXIs, we present evidence that "intervening against the Fed" mitigates market disturbances when US monetary policy changes suddenly. When the Fed hikes rates unexpectedly and the country sells the US dollar and buys domestic currency to counteract the surprise, the domestic currency does not depreciate against the US dollar, the currency risk premium and portfolio flows remain stable, and stock prices for both firms with and without US dollar debt remain unchanged. These results suggest that FXIs offset the spillover of US monetary policy via the balance sheet channel: it mitigates a depreciation of the exchange rate and consequently higher debt repayments for firms with US dollar debt.[3] Moreover, FXIs are also successful in preventing a rise in the currency risk premium and portfolio outflows, further stabilizing overall markets and offsetting the spillover of US monetary policy through a financial market channel. Our results highlight the importance of taking FXIs into account when analyzing monetary policy spillovers. When we pool no-intervention and intervention events, it is difficult to find evidence in favor of US monetary policy spillovers, as this pooling masks important heterogeneity in the effects of US monetary policy abroad.

To credibly counter-intervene against the Fed to stabilize their exchange rate, countries need to have accumulated a sufficient level of reserves. In fact, since the Asian financial crisis in the 1990s, emerging market economies have built up reserves to hedge against rollover risk (Bianchi et al., 2018).[4] When reserve levels are low, central banks' attempts to stabilize the exchange rate and consequently the economy may be unsuccessful as they lack the credibility to do so in the future (Fanelli and Straub, 2021). Empirically, we find evidence that countries that have a large share of FX reserves are more effectively counter-intervening against the Fed relative to their counterparts. This result suggests that the buildup of reserves over the last decades could have dampened the role of US monetary policy spillovers if countries had counter-intervened.

While FXIs may prevent a depreciation in response to a contractionary monetary policy shock, they do not mitigate the negative demand effects on imports originating from the US. The benefits of the contractionary monetary policy shock through the expenditure-switching channel for exporters may therefore vanish. The costs, however, through reduced demand, remain the same. Consequently, FXIs may harm exporters through a less depreciated exchange rate. Indeed, we find suggestive evidence that when countries do intervene, the effects of intervening against the Fed are negative for exporting firms, especially those without US dollar debt. This presents a tradeoff for the central bank. Intervening against the Fed helps stabilize risk premia, portfolio flows, and the stock prices of firms, especially those with US dollar debt—but it can harm exporters by appreciating the domestic currency. A large exporting country may therefore choose not to intervene, prioritizing export competitiveness, while countries with significant US dollar debt exposure may be more inclined to intervene.

Our identification strategy exploits the deviation from an estimated FXI policy rule and extracts the unexpected component of FXI in an attempt to use exogenous variation. However, this approach is not free of limitations, since it is difficult to find a perfect instrument in a cross-country panel setting, as discussed in Fratzscher et al. (2019).

[2] Under dominant currency pricing (Gopinath et al., 2020), export quantities are not expected to increase in the short-run, but revenues in domestic currencies are, also with positive implications for the stock price of exporters.

[3] One potential concern could be that firms use derivatives to hedge their exchange rate risk. While we do not have derivative data available, recent studies suggest that only a small fraction of firms hedge their FX risk with derivatives; see, e.g., Casas et al. (2022). Moreover, we introduce a statistical approach that proxies whether firms hedge their exchange rate risk, and we do not find that those firms affect our results. See also discussion in Appendix A.

[4] For example, the Swiss National Bank has sold the franc to counter the overvaluation due to its safe-haven status and boost the export industries (Jordan, 2017). Chinese foreign reserves increased from 733 billion US dollars in July 2005 to 3.99 trillion US dollars in June 2014 (Das, 2019).

There can be other confounding factors that affect both our dependent variable (stock returns and the exchange rate) and our independent variable (FXI) concurrently and introduce an omitted variable bias to our estimates. It is still possible that unwarranted market disturbances or negative news about the economy that are not captured in the FXI rule may affect stock prices and the exchange rate in response to US monetary policy, while at the same time inducing the central bank to intervene in the FX market.

While we acknowledge this limitation and encourage the subsequent literature to further improve upon this identification challenge, it is unlikely that this omitted variable fully explains our results. The exchange rate tends to depreciate systematically in "bad times" against the US dollar (Hassan et al., 2023). When the negative shock hits the economy and stock prices fall, one would expect the central bank to sell the US dollar to prevent the depreciation of the exchange rate. However, we find that selling the US dollar against the domestic currency is rather associated with an appreciation of the domestic currency and an increase in stock prices relative to the no-intervention case. This renders it unlikely that unobservable confounders are driving the relationship between FXIs and those outcome variables.

2. Literature review

Our paper contributes to the FXI literature and to how US monetary policy spills over to other countries. Our novel event study approach exploits daily firm-level stock prices across several countries, as well as exchange rates combined with FXI data and US monetary surprises purely identified by a high-frequency approach, see, e.g., Nakamura and Steinsson (2018).

A large empirical literature studies the effect of FXIs on exchange rates (Dominguez and Frankel, 1993; Dominguez, 2003; Dominguez et al., 2013; Adler et al., 2019; Fratzscher et al., 2019, 2023; Hofmann et al., 2019; Blanchard et al., 2015; Kuersteiner et al., 2018; Fatum and Hutchison, 2010).[5] However, none of these papers study the effect of FXIs on stock prices across the firm distribution, nor do they study the interaction between FXIs and US monetary policy.[6]

We also contribute to the literature on US monetary policy spillovers and the global financial cycle (Rey, 2015; Kalemli-Özcan, 2019; Miranda-Agrippino and Rey, 2020; Boehm and Kroner, 2025).[7] There is vast empirical evidence of the spillover of US monetary policy to country-level equity markets (Zhang, 2022; Wiriadinata, 2021; Dedola et al., 2017), exchange rates (Gürkaynak et al., 2021; Roussanov and Wang, 2023; Eichenbaum and Evans, 1995; Faust et al., 2007; Anderson et al., 2003), and interest rates (Timmer, 2018; Zhang, 2022). To identify the US monetary transmission channels, Wiriadinata (2021) uses foreign currency debt data, while Zhang (2022) uses currency invoicing shares. However, as discussed in Section 1, using cross-country heterogeneity faces several identification issues due to the presence of country-specific confounding factors and the masking of within-country heterogeneity.[8]

In contrast, we study the spillovers of US monetary policy on stock prices using a cross-section of firms' stock returns. Using daily stock-price data combined with high-frequency US monetary policy shocks allows us to understand the channel through which US monetary policy spills over to other countries. Surprisingly, at least to the best of our knowledge, we are the first to study the cross-section of stock price reactions to US monetary policy in an international context.[9] The advantage of using firm-level heterogeneity in stock price responses is that it allows us to identify the channels of US monetary policy more cleanly at a high frequency and only exploit differences across firms, controlling for time-variant country-specific observed and unobserved heterogeneity.[10]

The majority of literature on international US monetary spillover aggregates high-frequency US monetary shocks to lower frequency and studies their implication on the exchange rate, capital flows, and/or real outcomes (Miranda-Agrippino and Rey, 2020). In contrast, we combine daily data on exchange rates, firm stock prices, and FXIs with US monetary surprises. There are several advantages to using daily data as dependent variables. FOMC meetings occur at irregular intervals within each year, and aggregating monetary surprises over each month or quarter can induce serial correlation in aggregate shocks and inconsistent estimates of aggregate impulse responses (Ramey, 2016; Anderson and Cesa-Bianchi, 2024). Moreover, the magnitude of high-frequency shocks is small. This complicates estimating impulse responses since exchange rates in the distant future are influenced by many other confounding factors. To minimize this "power problem", we study the response of daily exchange rates and stock prices, which move contemporaneously with monetary shocks (Nakamura and Steinsson, 2018).

We also relate to the literature that studies how central banks respond when the US monetary policy tightens. In theory, the central bank should let the exchange rate depreciate (Friedman, 1953; Mundell, 1957; Fleming, 1962). However, in practice, many central banks are reluctant to let the exchange rate fluctuate due to a fear of floating (Calvo and Reinhart, 2002), as the evidence for the benefits of a depreciated currency for boosting exports remains elusive (Gopinath et al., 2020). Our results show that the monetary policy-induced depreciation does not benefit exporters as the positive expenditure switching effect is offset by negative demand effects, while at the same time harming firms that have US dollar borrowing.

The IMF integrated policy framework (Basu et al., 2020) studies the interaction between monetary policy and FXIs from a theoretical perspective. They show that after an adverse shock to the foreign appetite for domestic currency debt, FXIs reduce the need for the policy rate to be increased and, in that sense, can enhance monetary autonomy. Yago (2024) shows that by intervening against the Fed, central banks can offset the inflationary pressure driven by a US monetary policy-induced

[5] Recent theoretical advances show that interventions can affect the exchange rate and enhance welfare. Gabaix and Maggiori (2015), Cavallino (2019), Amador et al. (2020), Fanelli and Straub (2021), Maggiori (2022), Itskhoki and Mukhin (2023), and Yago (2024) study FXIs under partial segmentation of home and foreign currency bond markets. Since international financial intermediaries have limited risk-bearing capacity, FXIs affect the exchange rate by changing their balance sheet composition. Hassan et al. (2023) show that policies that appreciate domestic currency when the marginal utility of world investors is high increase the market value of firms and stabilize the country's wealth. Hence, small countries optimally choose to stabilize their currencies relative to the US dollar.

[6] Roussanov and Wang (2023) show that global FX dealers buy US dollars in response to contractionary monetary policy shocks, potentially representing the counterpart of the foreign central banks that sell the US dollar in response to contractionary monetary policy shocks.

[7] Akinci and Queralto (2024), Aoki et al. (2021), and Gopinath and Stein (2021) study US monetary policy theoretically and consider corporate or bank balance sheet risk when firms issue debt denominated in dollars.

[8] An exception is Morais et al. (2019), who match loan-level bank lending data with firm-level balance sheet data and study the international risk-taking channel of US monetary policy. While they use monthly loan data, we use daily data on FXIs and firm-level stock prices, as discussed below.

[9] A vast literature studies US firm-level equity returns in response to US monetary policy but abstracts away from foreign firms (Gorodnichenko and Weber, 2016; Ai et al., 2022; Ozdagli and Velikov, 2020; Chava and Hsu, 2020; Gürkaynak et al., 2022).

[10] The caveat of using stock price data is that we are only able to study public firms, and we cannot see whether a reduction in stock prices eventually materializes in changes in real outcomes, such as employment, domestic revenue, and profitability of those firms, see, e.g., Rodnyansky (2019).

currency depreciation. This provides a rationale for why intervening against the Fed can be optimal.[11]

Finally, our paper is also related to the literature studying financial channels of US monetary policy. In particular, US monetary policy shocks can alter investors' risk perception (e.g., Bernanke and Kuttner (2005) and Bruno and Shin (2015a,b)) and global stock market volatility (e.g., Miranda-Agrippino and Rey (2020) and Bauer et al. (2023)), the yield curve (e.g., Nagel and Xu (2024), Hanson and Stein (2015) and Timmer (2018)), shift portfolio flows (e.g., Converse and Mallucci (2023) and Converse et al. (2023)), and widen the currency risk premium (e.g., Lustig et al. (2011, 2014) and Kalemli-Özcan (2019)).

3. Data

3.1. Sources

We combine data from several sources. Our sample period is between 2000 and 2019, during which the data on US monetary shocks and corporate balance sheets are available.[12]

First, we collect data on sterilized FXIs based on Fratzscher et al. (2019) and Adler et al. (2025). We use publicly available databases on central bank websites and the FRED database.[13] Some countries do not publicly disclose the data due to the secret interventions, in which case, we individually contacted the central banks to be granted access to the data. We then restricted the sample countries based on the following criteria: first, to use as much high-frequency data as possible, we only use daily intervention data and exclude countries and periods where only monthly or quarterly data is available. Second, since we study interventions against US monetary shocks, our sample only comprises countries that intervened against the US dollar multiple times during the sample period.[14] The following 13 countries have available data and satisfy the above criteria: Argentina, Australia, Brazil, Chile, Colombia, Costa Rica, Georgia, Hong Kong, Japan, Mexico, Peru, Switzerland, and Turkey.[15] We make the FXI data publicly available for future researchers to build on this paper's findings. One limitation is that the intraday data on FXIs is difficult to obtain. However, even if we use daily data, our result shows that FXIs have persistent effects on the exchange rate and stock market.[16]

To address the endogeneity concern of monetary policy, we use the high-frequency change in the Fed funds rate (FFR) identified by Nakamura and Steinsson (2018), which was subsequently updated by Acosta (2023). They estimate the changes in Fed funds futures in a 30-minute window around the FOMC announcement. We obtain daily data on the spot exchange rate and stock price from Thomson Reuters Datastream. Since Datastream reports the closing exchange rate at 4 p.m. London time and the FOMC announcements are made around 2 p.m. Washington time (7 p.m. London time), we moved the reported dates of exchange rates by one date forward so that the exchange rate on each date is reported after the FOMC announcement on the same date. Since the end-of-date stock price is released at different times in different time zones, we adjusted the reported dates for the stock price depending on whether the stock price is released before or after the FOMC announcement.

We use corporate balance sheet data on the Capital IQ platform provided by S&P Global Market Intelligence. The advantage of Capital IQ is that it provides information on the currency denomination of debt, which is not available in other databases such as Worldscope, Compustat, and Orbis. Its Capital Structure database provides detailed information on each debt instrument held by each firm, including the principal amount due, repayment currency, and maturity. For example, Agrometal S.A.I., a manufacturing firm in Argentina, had a total outstanding debt of 5.6 million dollars on December 31, 2015. Among them, 2.2 million dollars are repaid in US dollars, and the remaining 3.4 million dollars are repaid in Argentine pesos. Hence, the share of dollar bonds over total bonds is 39 percent. Capital IQ has provided annual data on corporate balance sheets since 2001. The sample is restricted to publicly listed firms, as the data on stock prices is available.

To measure investors' risk aversion, following the literature (Bekaert et al., 2013; Bauer et al., 2023), we use the VIX index, which is calculated by the Chicago Board Options Exchange based on the implied volatility in S&P 500 stock index. We construct the currency risk premium using daily exchange rates and interbank rate data available in Bloomberg. The formal definition of the currency risk premium is provided in Appendix B. In addition, we collect the 10-year government bond yield data from Global Financial Data to measure the risk-free rate used for discounting future cash flows. Finally, to study the portfolio rebalancing channel, we use the EPFR equity fund flows dataset, which is available as a country-time panel at a weekly frequency. Investor flows are measured as end investors' purchases and redemptions of shares in mutual funds and ETFs from EPFR Global over the period since 1997, following the approach in Jotikasthira et al. (2012), Converse and Mallucci (2023) and Converse et al. (2023). The EPFR flow dataset is a country-time panel at a weekly frequency.[17]

We complement the data using a variety of sources. Firm-level data on exports, international sales, and international assets, as well as the incorporation date, are available on Worldscope. To measure firms' reliance on intermediate imports in their production, we use sector-level data on import content of exports in OECD input–output tables, following Rodnyansky (2019). The limitation is that it is difficult to obtain firm-level data on the invoicing currency of exports and imports for many countries. Hence, this paper focuses on the effect of FXIs on firms with dollar debt. Appendix B provides details on the data cleaning procedure and selection criteria for our sample of firms.

Finally, for country-level characteristics, we collect the data on monthly policy rates from the IMF Monetary and Financial Statistics. The data on GDP, inflation rate, trade balance, and unemployment rate are retrieved from the World Bank database and the IMF World Economic Outlook.

[11] Kalemli-Özcan (2019) shows that monetary policy divergence vis-à-vis the United States has larger spillover effects in emerging markets than in advanced economies. Domestic monetary policy is ineffective in mitigating this effect, as the pass-through of policy rate changes into short-term interest rates is imperfect.

[12] Appendix A shows that the result is robust even after excluding special periods, including the global financial crisis and zero lower bound on the interest rate.

[13] We focus on central banks' direct purchases and sales of the US dollar. One may think the signaling channel of FXIs is driving the results, since dollar sales may signal the central bank's concern about the weakness of the domestic currency, which is correlated with future monetary tightening. To address this possibility, Appendix A controls for government bond yield, which reflects future monetary policy expectations, and confirms our results on stock return and exchange rate remain robust. This confirms that the monetary policy expectation channel is not responsible for our results.

[14] We focus on US monetary shocks since a limited number of countries disclose daily FXI data against the euro and Japanese yen.

[15] To address the possibility that the result is driven by a particular country, in Appendix A, we exclude each country from the regression and show the results on stock prices and exchange rates are robust.

[16] Kuersteiner et al. (2018) study intraday intervention data in Colombia between 2001 and 2012. However, intraday data is unavailable for the other 12 countries in our sample. See also Dominguez (2003) and Dominguez et al. (2013) for analysis using intraday intervention data.

[17] We thank Nathan Converse for his help with the data construction. EPFR provides a comprehensive dataset on fund-level investor flows into ETFs and mutual funds. Using this data, the flows to individual countries are estimated based on the methodology described in Jotikasthira et al. (2012). Flows into fund f during each week are allocated to country c using fund f's reported portfolio allocation in country c at the end of the previous calendar year. The data is not available at a daily frequency. Each observation is reported on Wednesday and corresponds to the flows over the seven days until that Wednesday. For example, the April 2 observation reports the flow from March 27 through April 2.

3.2. Summary statistics

This section describes the summary statistics for FFR shocks and FXIs, and the number of sample firms across countries. The detailed tables are available in the appendix.

Table A.1 provides summary statistics for FFR shocks and changes in the exchange rate and stock price. Our sample consists of 90 FOMC announcement dates between 2000 and 2019.[18] Row (1) shows the summary statistics for the FFR shock estimated by Nakamura and Steinsson (2018). The FFR shock is defined as the change in market expectations of the Fed funds rate over the remainder of the month in which FOMC meetings occur. The shock is in terms of basis points, and a positive value implies a tightening surprise by the Federal Reserve. As Nakamura and Steinsson (2018) discuss, the magnitude of FFR shocks estimated by the high-frequency method is small: the standard deviation is 1.81 basis points. This "power problem" makes it difficult to estimate the effect on real economic outcomes, such as firm-level investment and economic growth, as the data is available only in quarterly or annual frequency and is affected by many unobservable confounding factors. Hence, we focus on the response of daily exchange rates and stock prices, which move contemporaneously with high-frequency monetary shocks.

Row (2) shows the summary statistics for the exchange rate depreciation, comparing before and after the FOMC announcement date. $e_{c,t}$ is the spot exchange rate at the end of date t in the country c. The exchange rate is defined as the value of the US dollar in terms of local currency, so that a higher $e_{c,t}$ implies the appreciation of the US dollar or depreciation of local currency. We take the change in the logarithm of the exchange rate from date $t-1$ to $t+1$. Similarly, row (3) shows the summary statistics for the percentage change in the stock price, comparing before and after the FOMC announcement. $p_{i,t}$ is the stock price of firm i at the end of date t. The stock price is denominated in local currency.[19] The standard deviations of exchange rate change and stock return are 0.72 percent and 3.47 percent, respectively.[20]

Table A.2 shows the frequency, amount, and sample period of interventions around the FOMC event dates in our sample countries. To consider the possibility that the effects of FFR shocks and FXIs accumulate over time, we consider a 5-day window after FOMC announcement dates.[21]

We first define buying and selling interventions so that central banks buy or sell the US dollar at least once between dates t and $t+5$, where t is the FOMC announcement date. Columns (1) and (2) report the frequency of buying and selling US dollar interventions. There is a large variation in the frequency of interventions across countries. For example, Argentina intervened 59 times by buying the US dollar and 45 times by selling it, out of a total of 90 FOMC event dates in our sample. In contrast, Switzerland never intervened, and Turkey intervened only once around the FOMC meetings. However, the interventions happened only outside the 5-day window around the meetings. To minimize the possibility that the interventions are affected by a myriad of other factors besides US monetary shocks, our sample does not count interventions that happened outside the 5-day window around FOMC meetings.[22]

Column (3) reports the frequency of counteracting interventions, which is the main focus of our analysis. We define counteracting interventions as follows: if the Fed funds rate increases on date t, central banks sell the US dollar at least once and never buy the US dollar between dates t and $t+5$, and vice versa when the Fed funds rate decreases. Since a higher Fed funds rate depreciates the local currency, central banks offset the depreciation by selling the US dollar and buying the local currency. Unless otherwise stated, we will use this definition of counteracting interventions throughout the regression analyses in this paper. However, our result is robust even if we adopt different definitions of counter-intervention or if we include all intervention dates in our sample.[23] Columns (1)–(3) show that not all interventions are counteracting interventions against the Fed. For example, Argentina intervened by buying the US dollar 59 times and by selling the US dollar 45 times, but only on 15 occasions did they intervene by counteracting the FFR shock. Focusing on these counter-interventions, we will study how the effects of FOMC announcements on the exchange rate and stock price are different when the central banks do and do not counter-intervene against FFR shocks. Columns (4) and (5) report the FXI volume in terms of millions of US dollars around the FOMC announcement dates. The mean and median FXI volumes across all countries are 57 million dollars and 15 million dollars, respectively.[24]

Column (6) reports the sample period when the FXI data are available. The availability of the FXI data depends on the country. In particular, our sample only includes countries and periods in which daily FXI data is available. We excluded periods in which only monthly, quarterly, or annual data are available. For example, daily intervention data in Switzerland is available only until 2001.

To study how long foreign exchange interventions last, we follow Fratzscher (2008) and Fratzscher et al. (2019) and define the two intervention days as being in the same episode if they are in the same direction and within a ten-trading-day (two-week) time window. Based on this definition, the mean and median lengths of intervention episodes are 10.7 and 7 days, respectively.

Table A.3 describes the sample of firms. The table shows the total number of firms and the number of firms that issued dollar debt at least once during the sample period in each country. Our sample consists of 4060 firms in total, out of which 261 firms (6%) have dollar debt. The average share of dollar debt over total debt across all firms is 66%, conditional on firms issuing a positive amount of dollar debt. The share of firms with dollar debt is relatively large in emerging economies, while most of the Japanese firms do not issue dollar debt as they borrow in Japanese yen (if we exclude Japanese firms from the sample, 14%

[18] Nakamura and Steinsson (2018) and Acosta (2023) report 151 FOMC sample dates between 2000 and 2019. We exclude 46 dates that are characterized by a zero change in the Fed funds rate. This is because our main focus is the unexpected change in the Fed funds rate and the central banks' counteracting intervention against FFR shocks. In Appendix A, we also include those dates in the sample and show that the result is robust. We also excluded 15 FOMC event dates with large FFR shocks so that our results are not affected by outliers (see footnote 20).

[19] Appendix A shows that, if the stock price is denominated in foreign currency, the stock price response to Fed tightening becomes larger since changes in exchange rates affect the valuation of stock prices. This implies that firms with dollar debt are riskier investment opportunities for foreign investors than those with local currency debt.

[20] We trimmed the top and bottom 5% of the FFR shocks and winsorized the top and bottom 5% of changes in the exchange rate and firms' stock prices in each country so that our results are not affected by outliers.

[21] For identification purposes, we focus on a 5-day window around FOMC announcements, since the estimates can be more affected by unobserved confounding factors in a longer time horizon. However, policymakers may be more interested in a longer-lasting effect, as the balance sheet and expenditure switching effects occur at a lower frequency. In Appendix D, we study this lower-frequency effect and find that the effects of US monetary shocks and FXIs on stock prices are persistent for six to eight weeks (1.5 to 2 months).

[22] To address the possibility that our result is affected by one particular country with either frequent or infrequent intervention, Appendix A excludes each country from the sample and shows that our result is robust.

[23] Appendix A defines the counteracting interventions differently: the central banks' average net sales of US dollars between dates t and $t+5$ is positive when the Fed funds rate increases at date t, and vice versa when the Fed funds rate decreases. Our main results are robust to using this alternative definition.

[24] To calculate the mean and median FXI volumes around FOMC announcement dates, we first take the average of absolute values of FXI volumes between date t and $t+5$ for each FOMC announcement date t. Next, we take the mean and median of implied FXI volume over FOMC announcement dates, conditioning that there was at least one intervention between date t and $t+5$.

of firms have dollar debt). As discussed in Appendix A, almost all of the sample firms with dollar debt do not have export revenues and therefore are not naturally hedged.

3.3. Identification of unexpected FXI

To elaborate on our identification strategy for FXIs, we estimate a central bank reaction function. The motivation is to extract the unexpected component of interventions that cannot be forecasted by FFR shocks, past exchange rate movements, FXIs before the FOMC events, and other macroeconomic characteristics. This is a popular approach to minimizing the endogeneity of interventions in the literature (Kearns and Rigobon, 2005; Ito and Yabu, 2007; Fatum and Hutchison, 2010; Kuersteiner et al., 2018; Fratzscher et al., 2019), similar to residuals from a monetary policy Taylor rule. The advantage of the deviation from the FX policy rule relative to deviations from a monetary policy rule is that FXIs vary on a daily level, while monetary policy decisions are usually only conducted every several weeks. Therefore, the high-frequency FXI rule approach more cleanly identifies surprises than the Taylor rule residual.[25]

All of our results are robust across various definitions or measures of FXIs. If we do not use indicators but instead adopt continuous measures for the raw volume (size) of FXIs, or if we do not take the surprise component by estimating the FXI rule and include all intervention dates in our sample, our key economic implications on exchange rates and stock prices do not change. However, we use the following definition of FXIs as a benchmark to better identify the direct effects of FXIs that are less likely to be confounded by other factors.

To do so, we consider the following FXI rule:

$$\widetilde{FXI}_{c,t} = \sum_c \beta_c (FFR_t \times \gamma_c) + \delta Z_{c,t} + \gamma_c + \epsilon_{c,t}. \tag{1}$$

$\widetilde{FXI}_{c,t}$ is the indicator for counteracting intervention in country c on FOMC announcement date t, as discussed in column (3) of Table A.2, as is standard in the literature.[26] $\widetilde{FXI}_{c,t}$ takes 1 if the Fed tightens unexpectedly on date t and the central banks intervene by selling the US dollar at least once, but never intervene by buying the US dollar between dates t and $t+5$. Similarly, $\widetilde{FXI}_{c,t}$ takes -1 if the Fed loosens unexpectedly on date t and the central banks intervene by buying the US dollar at least once but never intervene by selling the US dollar between dates t and $t+5$. Otherwise, $\widetilde{FXI}_{c,t}$ takes zero. FFR_t is the FFR shock in terms of basis points, and γ_c is the fixed effect for each country c. Their interaction $FFR_t \times \gamma_c$ captures the differential propensity to intervene against FFR shocks across countries. $Z_{c,t}$ is the set of controls, including the trend and standard deviation of the exchange rate and the dummy for FXIs before the FOMC event date, as well as the macroeconomic variables (one-month lagged policy rate, one-year lagged GDP, CPI inflation rate, trade balance over GDP ratio, and unemployment rate), and the interaction of macroeconomic variables with FFR shock.[27] For past exchange rate movement, we took the percentage change and standard deviation of the exchange rate between dates $t-1$ and $t-5$, where t is the FOMC event date. For past interventions, the dummy takes 1 if the average net purchase of US dollars between dates $t-1$ and $t-5$ is positive, -1 if the net purchase is negative, and zero if there are no interventions.

We follow the previous empirical literature on FXIs regarding the estimation of an FXI policy rule. Following Fratzscher et al. (2019), we control for the past exchange rate trend and volatility and past interventions, as they can affect the central bank's decision to intervene. Moreover, based on Fatum and Hutchison (2010), we also control for macroeconomic variables, such as GDP and trade balance, since countries with different macroeconomic conditions may adopt different interventions. We use lagged macroeconomic variables to remove the simultaneity bias. We also account for the interaction between the FFR shock and macro variables, since countries with different macroeconomic characteristics can respond heterogeneously to the FFR shock. We include the country fixed effects, γ_c, to control for the difference in average exchange rate trends in each country.

The predicted counter-intervention from estimating the reaction function (1) can be interpreted as the expected component of counter-interventions, in other words, the average response of FXIs to FFR shocks, past exchange rates, interventions, and macroeconomic conditions. The residual, or the deviation from the FXI rule, can be interpreted as the unexpected component of FXIs. We exploit this residual as the exogenous surprise component of FXIs.

The appendix provides a graphical illustration of the results from estimating the FXI policy rule. Fig. A.1 shows the result for a variance decomposition from estimating Eq. (1). Our estimates show that 24% of the variation in counter-intervention can be explained by the set of controls ($R^2 = 0.24$), while the remaining 76% is unexplained. This low R-squared implies that a large part of FXIs cannot be predicted by the FFR shock, past exchange rates, or interventions. If we further decompose the controls, 16% can be explained by the FFR shocks, 2% by macroeconomic variables (policy rate, GDP, CPI inflation rate, trade balance over GDP ratio, unemployment rate), 30% by the interaction between the FFR shock and macro variables, 35% by past interventions, and 17% by country fixed effects. The contribution of past exchange rate trends and volatility is almost zero, so it is not displayed in the figure. This is consistent with Fratzscher et al. (2019), who show that past exchange rates have very limited explanatory power for FXIs.

Having shown that the variation of interventions is difficult to predict, we will use the residual from estimating Eq. (1) as an unexpected component of counter-interventions. To simplify the interpretation of results in later sections, we define an unexpected counter-intervention dummy, $FXI_{c,t}$. If the residual from estimating Eq. (1) is greater than its median in absolute value, $FXI_{c,t}$ takes one, and central banks counter-intervene unexpectedly against the Fed. Otherwise, $FXI_{c,t}$ takes zero, and central banks do not counter-intervene unexpectedly. Although we use this indicator as the benchmark, we emphasize again that our results are robust even if we do not take dummies but instead the continuous measure for the raw volume of FXIs.

Fig. A.2 shows a simple graphical example of this identification methodology. Panel (a) shows an example of unexpected US monetary tightening: on November 15, 2000, the Fed tightened unexpectedly, and the Reserve Bank of Australia intervened by selling the US dollar. The counter-intervention dummy $\widetilde{FXI}_{c,t}$ takes 1. The predicted value and residual from estimating the linear probability model (1) are around 0.2 and 0.8, respectively. This implies that the market expected that there was a 20% probability that the central bank would intervene against the Fed's tightening shock by selling the US dollar, but an 80%

[25] This approach is conceptually similar to the estimation of the Taylor rule residual for monetary policy (Taylor, 2009; Maddaloni and Peydró, 2011; Dell'Ariccia et al., 2017). In a similar vein, Christiano et al. (1996, 1999) estimate a structural vector autoregression (VAR) model to identify monetary policy shocks, and Blanchard and Perotti (2002) estimate a VAR model to identify fiscal policy shocks.

[26] We use an indicator variable for FXIs following the literature on FXI reaction function: Fatum and Hutchison (2010) and Fratzscher et al. (2019) (Appendix II) define a (0,1) indicator for intervention and no intervention, and Ito and Yabu (2007) define a (0,1,−1) indicator for buying and selling dollar interventions. Ito and Yabu (2007) use an FXI indicator instead of volume because the FXI volume is determined within the day depending on intraday exchange rate movement, but intraday interventions are not disclosed. By using the indicator variable, they mitigate this endogeneity concern caused by our inability to estimate the intraday reaction function.

[27] We use a monthly policy rate since the daily policy rate is not available for all sample countries. In Appendix A, we use daily policy rates in countries with available data and show that our result is robust.

probability that the central bank would not intervene. This large residual implies that the interventions were mostly unexpected. Similarly, panel (b) shows the example of unexpected US monetary easing: on March 22, 2005, the Fed delivered an accommodating monetary policy shock, and the Central Bank of Argentina intervened by buying the US dollar. The counter-intervention dummy $\widehat{FXI}_{c,t}$ takes −1. The predicted value and residual from estimating the linear probability model (1) are around −0.39 and −0.61, respectively. This implies that the market expected that the probability that the central bank would not intervene against the US monetary easing shock was more than 60% (the residual is 0.61 in absolute value). This suggests that the degree to which the interventions are unexpected can be measured using the absolute value of the residual from the estimating Eq. (1). In the baseline analysis, we define the interventions as unexpected if the residual is larger than its median in absolute value. Our result is robust even if we use different criteria for defining unexpected counter-intervention or use the continuous size of FXIs and include all intervention dates without taking the unexpected component.

The advantage of our identification approach is that, by estimating the residual from an FXI policy rule, we can extract the surprise component of FXI and study its effect on the exchange rate. Our approach, however, is not without limitations. It is difficult to find a perfect instrument since unobserved confounders, such as investors' sentiments or expectations, can affect FXIs, exchange rates, and stock returns at the same time.

Future literature is encouraged to explore other identification strategies. One possible avenue for research is the narrative approach. Literature on monetary policy, which was pioneered by Friedman and Schwartz (1963) and later developed by Romer and Romer (1989), uses historical FOMC records or news articles to identify monetary policy decisions that were not driven by inflation or output.[28] The narrative approach is powerful but challenging to apply to FXI, since intervention decisions are often made in secret and not publicly disclosed.[29] Future availability of FXI data and firm-level currency exposure would improve the accuracy of our estimates and enable more comprehensive analyses of the effect of FXI on the exchange rate and the stock market.

4. Empirical strategy

Our empirical strategy relies on a high-frequency event-study approach that examines the performance of equities and exchange rates around FOMC meetings. The event study approach has the advantage that the market reaction on FOMC dates is likely due to monetary policy itself, rather than other confounding factors that could influence equity prices or the exchange rate. For instance, in a simple time-series regression in which quarterly outcome variables are regressed on US monetary policy (shocks), it is more difficult to identify the causal effect of monetary policy as many confounding factors could be the reason for the market reaction that is not due to monetary policy itself. If the monetary policy shock is completely exogenous, the coefficient may not be biased, but aggregating high-frequency monetary policy shocks to the quarterly level may cause a "power issue", similar to a weak instrumental variable problem in two-stage least squares regressions.

We start with event-study local projections (Jordà, 2005) by estimating the following sequence of regressions across FOMC dates:

$$y_{i(c),t+h} - y_{i(c),t-1} = \beta_h \mathrm{FFR}_t + \mathbf{X}\delta_x^h + \alpha_{i(c)}^h + \epsilon_{i(c),t}^h, \quad \forall h = \in [-5, 5] \tag{2}$$

where FFR_t is the FFR shocks developed by Nakamura and Steinsson (2018) and Acosta (2023). The shock is defined as the change in the Fed funds futures rate in a 30-minute window around the FOMC announcement. FFR_t is in terms of basis points, and a positive FFR_t represents the unexpected increase in the Fed funds rate. $y_{i(c),t+h}$ is the logarithm of stock price of a firm i based in the country c, h days after the FOMC meeting. $\alpha_{i(c)}^h$ is a firm fixed effect. The standard errors are always double-clustered at the firm and event date level to account for correlation at the same firm and time. β_h is the effect of the FFR shock on the equal-weighted stock price h days after the FOMC meeting. Eq. (2) is informative about the spillover effects of US monetary policy across all countries in our sample and across firms. $\beta < 0$ for each $h \geq 0$ implies that a surprise tightening of US monetary policy reduces stock prices abroad h days after the meeting. Then we can estimate Eq. (2) for the country-FOMC date subsamples with and without unexpected counter-intervention ($FXI_{c,t} = 1$ and $FXI_{c,t} = 0$). For the subsample $FXI_{c,t} = 1$, $\beta = 0$ implies that monetary policy does not spill over negatively to countries' equally-weighted stock price index h days after the meeting if they counter-intervene against the Fed. $\mathbf{X}$ is a set of controls that include the one-year lagged export intensity, total asset, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interactions with FFR shock. The interaction of the Fed funds rate with the firm-level characteristics ensures that the differential responsiveness across firms to changes in interest rates is not responsible for the identified effect of the balance sheet channel.

We consider two modifications to evaluate the effect of FXIs. First, we introduce an interaction term between the Federal funds rate shock and dollar debt:

$$\begin{aligned} y_{i(c),t+h} - y_{i(c),t-1} = {} & \gamma_h \mathrm{FFR}_t \times USD_{i(c),y-1(t)} + \mathbf{X}\delta_x^h \\ & + \alpha_{i(c)}^h + \alpha_{c,t}^h + \epsilon_{i(c),t}^h, \quad \forall h = \in [-5, 5] \end{aligned} \tag{3}$$

γ_h can be interpreted as a state-dependent effect of the monetary policy shock, where the state is whether firms have US dollar debt or not (Cloyne et al., 2023). $USD_{i(c),y-1(t)}$ is a dummy if a firm has had US dollar debt in the previous year before the FOMC meeting. Appendix A shows that our results are robust even if we do not take a dummy but instead use a continuous measure for dollar debt. $\alpha_{c,t}^h$ is a country-time fixed effect and $\alpha_{i(c)}^h$ is a firm fixed effect. γ captures the balance sheet channel of US monetary spillover. $\gamma < 0$ for each $h \geq 0$ implies that a surprise tightening of US monetary policy reduces the stock price of firms with dollar debt relative to those without dollar debt. Note that the inclusion of the country-time fixed effects controls for all time-varying observed and unobserved characteristics, and only exploits the effects across firms within a country-time dimension. The inclusion of the fixed effects also implies that all other variables that are spanned by the fixed effects are collinear and cannot be estimated, e.g., FFR_t itself.

Next, we introduce an interaction term between the Federal funds rate shock and the intervention:

$$\begin{aligned} y_{i(c),t+h} - y_{i(c),t-1} = {} & \beta_h \mathrm{FFR}_t + \Omega_h FXI_{c,t} + \gamma_h \mathrm{FFR}_t \times FXI_{c,t} \\ & + \mathbf{X}\delta_x^h + \alpha_{i(c)}^h + \epsilon_{i(c),t}^h, \quad \forall h = \in [-5, 5] \end{aligned} \tag{4}$$

γ captures the effect of FXIs mitigating the effect of US monetary spillover. $\gamma > 0$ for each $h \geq 0$ implies that the decline in stock price is smaller with FXIs than the case without FXIs. In contrast to standard local projections, we also consider $h < 0$ in the spirit of an LP-DID proposed by Dube et al. (2025). One difference between the LP-DID and the standard DID is that a sequence of regressions is estimated for each h. This has the advantage that β_h is unaffected by the choice of the number of lags and leads included. Moreover, the LP-DID avoids several other problems compared to estimating a difference-in-differences specification with two-way fixed effects, see, e.g., Callaway and Sant'Anna (2021) and Goodman-Bacon (2021) among many others.

For the difference-in-differences estimator to be unbiased, we require the parallel trend assumption to be satisfied—that is, absent a shock, treated and control firms would have evolved the same way.

[28] See Romer and Romer (2023) for developments of the narrative approach and its application to recent data.

[29] An exception is Naef (2024), who hand-collects confidential daily reports on intervention decisions by the Bank of England from 1987 to 1992 to identify the surprise component of FXI.

While it is not possible to test this assumption, as the counterfactual post-FOMC behavior without the shock is unobservable, we can test whether there are differential pre-trends before the shock. Estimating β_h for $h < 0$ allows us to test whether there is a violation of the parallel trend assumption.

Recent literature has argued that DID designs are likely to be biased in the presence of a staggered DiD approach, as already treated units can act as effective comparison units (Baker et al., 2022). Note that this is not a concern in our setting as we set $h = \in [-5, 5]$, covering only a window of 10 days, which prevents overlapping observations and staggered treatment, as FOMC meetings only occur approximately every six weeks. The concern would be that firms with US dollar debt are treated for one FOMC meeting but not for the next, but would still be treated as comparison units for the next one.

We start by estimating Eqs. (2) and (3) separately for FOMC meetings when country c counter-intervenes in the FX market after the FOMC meeting, based on the definition in Section 3. Focusing on no-counter-intervention events allows us to test the degree and channels of US monetary policy spillovers if countries do not intervene against the Fed. Instead, focusing on counter-intervention events allows us to test whether US monetary policy spills over to firms' stock prices if a country intervenes against the Fed. In particular, if we can reject the null hypothesis that $\beta = 0$ in Eq. (2), the data favors the alternative hypothesis that there exists a spillover effect of US monetary policy. If we cannot reject $\beta = 0$, there are likely no spillover effects. In Eq. (3), the null hypothesis is that firms with US dollar debt are not differentially affected by a US monetary spillover, while the alternative hypothesis is that they are differentially affected. The alternative hypothesis, $\gamma < 0$, can be interpreted as a US monetary policy-driven balance sheet channel of depreciation.

The disadvantage of splitting the data into intervention and no-intervention events is that we cannot test whether β and γ are statistically different for intervention and no-intervention events. It is possible that when estimating the equations separately, we can reject the null hypothesis in one subsample but not in the other, yet due to large standard errors, the two situations are not different from each other in a statistically significant manner.

We therefore refine our regression equation by including the counter-intervention dummy specifically in the regression equation instead of splitting between situations when $FXI_{c,t}$ is either zero or one:

$$\begin{aligned} y_{i(c),t+h} - y_{i(c),t-1} = {} & \theta_h \mathrm{FFR}_t \times USD_{i(c),y-1(t)} \times FXI_{c,t} \\ & + \gamma_h \mathrm{FFR}_t \times USD_{i(c),y-1(t)} \\ & + \mathbf{X}\delta_x^h + \alpha_{i(c)}^h + \alpha_{c,t}^h + \epsilon_{i(c),t}^h, \quad \forall h = \in [-5, 5] \end{aligned} \tag{5}$$

In Eq. (5), the coefficient γ has the same interpretation as in the equation without the triple interaction (Eq. (3)) when estimating in the sample of no-intervention: the relative performance of firms with US dollar debt in response to a contractionary US monetary policy shock when the country does not counter-intervene. A negative coefficient implies that firms with US dollar debt underperform relative to those with US dollar debt. The triple interaction coefficient θ measures the marginal effect of FXIs for firms with US dollar debt in response to a contractionary monetary policy shock. A positive coefficient θ implies that FXIs lead to relatively higher stock prices for firms with US dollar debt in response to a contractionary monetary policy shock, compared to a counterfactual under which the central bank does not counter-intervene. θ can therefore be interpreted as the extent to which FXIs mute the US monetary policy-induced balance sheet channel of exchange rate depreciation, while γ measures the extent of the balance sheet channel without interventions. Summing θ and γ can be equivalently interpreted as Eq. (3) for $FXI_{c,t} = 1$ as the balance sheet channel when countries intervene. $\theta + \gamma = 0$ implies FXIs entirely mute the balance sheet channel.

Note that estimating the relative stock market response of having US dollar debt in response to the monetary policy shock allows us to saturate the regression specification with country-time fixed effects ($\alpha_{c,t}$). Country-time fixed effects control for time-variant observed and unobserved characteristics at the country level, such as the movement of the exchange rates or the effect on the average stock price around the US FOMC meetings. The inclusion of country-time fixed effects implies that the effect of FFR_t is not identified, as it is collinear with the fixed effects (note that $\alpha_{c,t}^h$ absorbs the terms FFR_t, $FXI_{c,t}$, and $\mathrm{FFR}_t \times FXI_{c,t}$). Hence, when controlling for time-variant observed and unobserved characteristics at the country level through country-time fixed effects, we can only make a relative statement about having US dollar debt. In an alternative specification, we remove the country-time fixed effects from the regression specification to evaluate the total effect of US monetary policy shocks for both firms with and without dollar debt, with the caveat of controlling for fewer potential confounding factors.

5. Results

We now turn to the empirical result. We study different transmission channels through which US monetary policy affects the foreign equity market and how FXIs modify these effects. First, Section 5.1 establishes that "intervening against the Fed" successfully mitigates the US monetary spillover via the balance sheet channel, exploiting firm-level balance-sheet heterogeneity for identification purposes. Next, Sections 5.2 and 5.3 study how US monetary policy and FXIs affect the exchange rate, risk premium, and portfolio flows. Section 5.4 shows that FXIs can harm exporters via the expenditure switching channel and discusses the central banks' tradeoff between export competitiveness and financial stability. Section 5.5 measures how much volume of FXIs is sufficient to offset the US monetary transmission to the stock market. Section 5.6 shows that FXIs are more effective in countries with large foreign exchange (FX) reserves. Finally, Section 5.7 provides robustness checks.

5.1. *Stock market transmission via the balance sheet channel*

We begin by estimating Eq. (2) separately for firms with and without US dollar debt, i.e., meaning that ($USD_{i(c),y-t(t)} = 1$ and 0). Table 1 shows the result, where panels (a) and (b) show the result with and without FXIs, respectively. We first study the case without FXIs ($FXI_{c,t} = 0$). Panel (a), columns (1) and (2) report the estimated β_1 coefficient for a subsample of firms with and without dollar debt, respectively, based in countries without FXIs. Column (1) shows that, if central banks do not counter-intervene against the Fed, an unexpected increase in the Fed funds rate reduces the stock price for firms with dollar debt in a statistically significant manner (6.6% in response to a 10 bp surprise hike).[30][31] However, if firms do not have dollar debt,

[30] As shown in Table A.1, the standard deviation of the monetary shock in our sample is very small (1.81 bp) since it is measured in a 30-minute window around the FOMC announcement. In fact, in the dataset of Nakamura and Steinsson (2018), there are only six FOMC announcement dates in our sample period (between 2000 and 2019) when the magnitude of the FFR shock is greater than or equal to 10 bp in absolute value. Hence, a 10 bp monetary shock can be interpreted as a large shock.

[31] One may think this decline in stock prices is driven by the portfolio rebalancing channel: a surprise US tightening lowers the US equity return and raises its expected returns, which induces US investors to shift funds to US assets. We would expect that this is not the main driver of our results. While the portfolio rebalancing channel can explain the overall decline in local stock returns, it is less likely to explain differential stock returns across firms, unless international investors disproportionately hold certain firms. One may potentially expect US investors to have a higher exposure toward larger firms, and the portfolio rebalancing channels are driven by large firms. To take this size effect into account, we control for firm size and its interaction with the FFR shock. This disentangles the balance sheet and portfolio rebalancing channels. Finally, Appendix A.1 exploits the debt maturity structure for identification. It is unlikely that international investors hold firms with differential maturity structure, even if they disproportionately hold firms with US dollar debt.

Table 1
Stock price: Baseline regression.

(a) Without intervention

Dependent variable:	ΔStock Price$_{i(c),t}$			
	Dollar debt	No dollar debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock$_t$	−0.660***	−0.094**	−0.097**	
	(0.117)	(0.045)	(0.045)	
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$			−0.314***	−0.259***
			(0.087)	(0.071)
R^2	0.093	0.032	0.031	0.083
N	1926	103,155	105,114	105,114
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

(b) With intervention

Dependent variable:	ΔStock Price$_{i(c),t}$			
	Dollar debt	No dollar debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock$_t$	−0.217**	−0.149***	−0.158**	
	(0.105)	(0.056)	(0.061)	
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$			−0.001	−0.033
			(0.042)	(0.035)
R^2	0.114	0.209	0.194	0.270
N	1258	9915	11,178	11,178
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

the decline in stock price is mitigated (0.9%). Next, to test if the difference in the response between firms with and without dollar debt is significant, we estimate Eq. (3) in countries without interventions. Column (3) reports the estimated β_1 and γ_1 coefficients for countries without interventions. The negative γ_1 coefficient shows that when the Fed funds rate increases, the decline in stock prices for firms with dollar debt is significantly larger than for those without dollar debt (3.1pp larger decline). Moreover, in column (4), we include the country-time fixed effect $\alpha^h_{c,t}$, which captures the time-varying observable and unobservable characteristics at the country level, and our result remains robust. Note that $\alpha^h_{c,t}$ absorbs the FFR$_t$ standalone term. These results suggest a strong negative balance sheet channel driven by US monetary tightening without FXIs.

In contrast, panel (b) conducts a similar exercise with FXIs ($FXI_{c,t} = 1$). Column (1) shows that, if central banks counter-intervene against the Fed's surprise hike, the decline of stock prices is mitigated even if firms have dollar debt (2.2% in response to a 10 bp surprise hike, in contrast to 6.6% in panel (a)). Columns (3) and (4) show the differential response of firms with and without dollar debt. The coefficient γ_1 is small and statistically insignificant, implying that if central banks counter-intervene against the Fed, the stock price response of firms with dollar debt is not different from those without dollar debt in a statistically significant manner. This suggests that FXIs can successfully mute the negative spillover of US monetary shocks via the balance sheet channel, disproportionately benefiting firms with dollar debt.

In addition, comparing column (3) of panels (a) and (b), the β coefficient on the FFR shock is negative and bigger in magnitude with FXIs ($\beta = -0.158$) than without FXIs ($\beta = -0.097$). This suggests the possibility that FXIs may harm firms without dollar debt. In Section 5.4, we test whether this negative effect can be explained by firms with international revenues and assets.

Next, to test whether the balance-sheet effect of FXIs is large in a statistically significant manner, we estimate Eq. (4) separately for firms with and without dollar debt. Columns (1) and (2) of Table A.4 report the estimated β_1 and γ_1 coefficients for firms with and without dollar debt, respectively. Column (1) shows that, if firms have dollar debt, a surprise hike in the Fed funds rate reduces the stock price without interventions (6.4% in response to a 10 bp hike), implying a negative balance sheet channel of US monetary policy. However, if central banks intervene, the decline in the stock price is mitigated by 4.5 bp, implying that FXIs can mitigate this balance sheet channel. In column (2), we conduct a similar exercise for firms without dollar debt and find that the effects of the US monetary shock and FXIs are small. Finally, in columns (3) and (4), we estimate the triple interaction (Eq. (5)) to study whether the effect of FXIs is greater for firms with dollar debt. The negative coefficient on $FFR_t \times USD_{i(c),y-1(t)}$ ($\gamma = -0.310$) implies a negative balance sheet channel due to US monetary policy spillovers. However, the coefficient on $FXI_{c,t} \times FFR_t \times USD_{i(c),y-1(t)}$ is positive ($\theta = 0.324$) and $\theta + \gamma = 0$ holds statistically. This result implies that FXIs entirely mute the balance sheet channel. The finding is robust even after including the country-time fixed effect (column 4). Note that $\alpha^h_{c,t}$ absorbs the terms FFR$_t$, $FXI_{c,t}$, and FFR$_t \times FXI_{c,t}$. This key implication is robust even if we do not take a dummy but instead use a continuous measure of FXIs, and even if we do not take a surprise component of FXIs but include all intervention dates in our sample.

To test whether the effect of FXIs is persistent over time, we estimate Eq. (2) over a 5-day window around the FOMC announcement. Fig. 2, panel (a) plots the estimated coefficient β_h for all $h = \in [-5, 5]$. The red and blue lines show the results for countries with and without FXIs, respectively. Without FXIs, a 10 basis point surprise increase in the Fed funds rate leads to an immediate decline in stock prices, and this accumulates up to nearly 10% after three days, suggesting that the balance sheet channel of US monetary shocks is persistent. However, if countries intervene, the effect of the US monetary surprise is smaller, and it disappears five days after the shock. Importantly, the near-zero

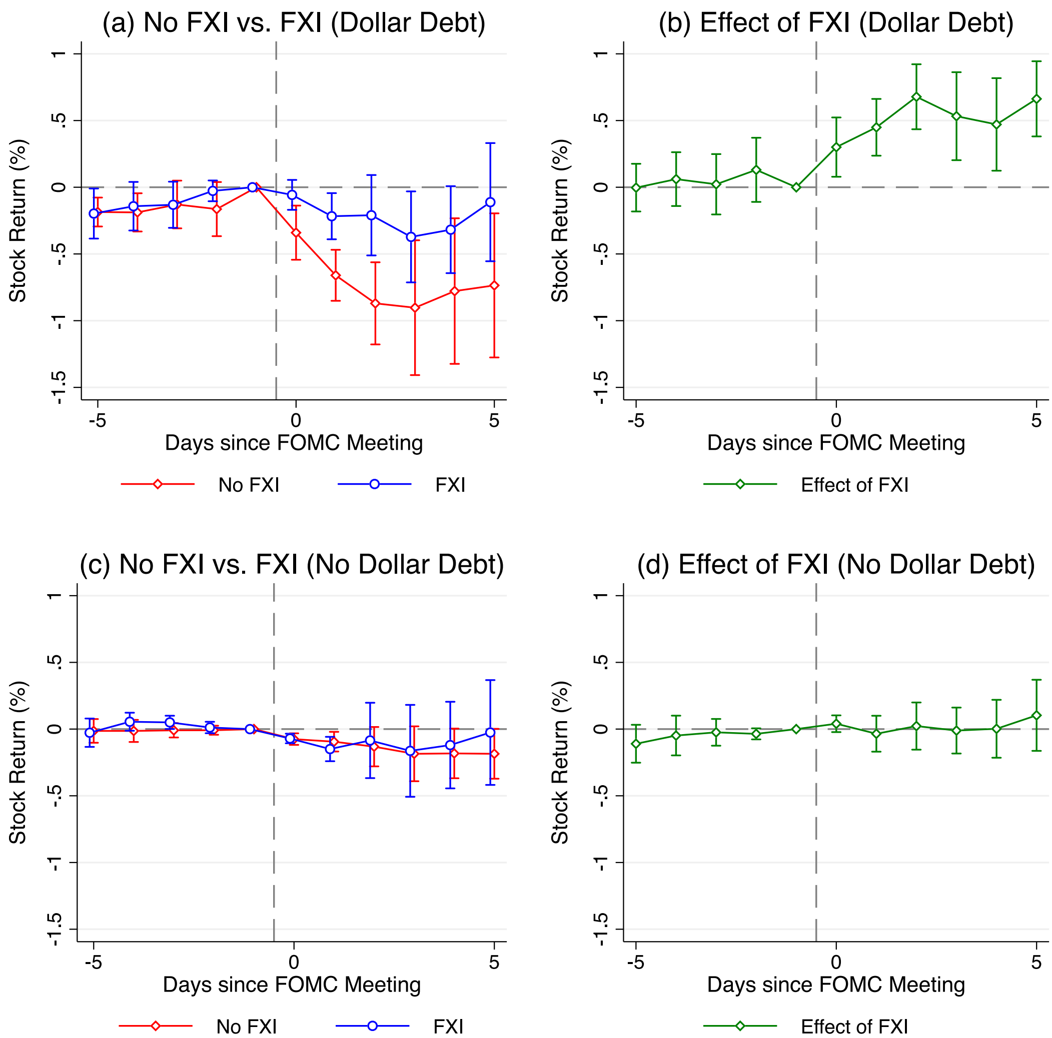

Fig. 2. Stock price: Difference-in-difference.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Eqs. (2) and (4) for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are double clustered at the firm and date level. The confidence interval is 90%. The red and blue lines show the results for countries with and without FXIs, respectively, and the green line shows the difference between them.

coefficients for $h < 0$ suggest that there is little difference in the pre-trends of stock prices in countries with and without FXIs, potentially suggesting that the post-FOMC differential response of stock prices is driven causally by FXIs. To test if the effect of FXIs is large in a statistically significant manner, in panel (b), we estimate Eq. (4) and plot the coefficient γ_h for all $h =\in [-5, 5]$. $\gamma_h > 0$ for $h > 0$ implies that FXIs can successfully mute the persistently negative balance sheet effects of US monetary shocks. In panels (c) and (d), we repeat a similar exercise for firms without dollar debt. The graph shows that the US monetary shock and FXIs have little effect on the stock price of firms without dollar debt. Finally, to compare the effect of FXIs on firms with and without dollar debt, we estimate Eq. (5) for all $h =\in [-5, 5]$. Fig. 3 plots the estimated coefficient θ_h. $\theta_h > 0$ for $h > 0$ suggests that the effect of FXIs is persistently greater for firms with dollar debt.

5.2. Risk premium and portfolio rebalancing

We have discussed how monetary policy affects stock prices through a balance sheet channel. However, the balance sheet channel may not be the only mechanism through which US monetary policy shocks affect foreign stock prices. In this section, we dig into how monetary policy affects stock prices through financial channels other than the balance sheet, including changes in the currency risk premium as well as portfolio rebalancing. In particular, US monetary policy shocks can alter investors' risk perception (e.g., Bernanke and Kuttner (2005) and Bruno and Shin (2015a,b)) and global stock market volatility (e.g., Miranda-Agrippino and Rey (2020) and Bauer et al. (2023)), the yield curve (e.g., Nagel and Xu (2024), Hanson and Stein (2015) and Timmer (2018)), shift portfolio flows (e.g., Converse and Mallucci (2023) and Converse et al. (2023)), and widen the currency risk premium (e.g., Lustig et al. (2011, 2014) and Kalemli-Özcan (2019)), while also influencing the US dollar. Each of these channels can, in turn, affect the pricing of foreign equities.

Below, we first test how US monetary policy affects those variables and whether FXIs mitigate them. We then show how the movements of those variables affect the pricing of foreign equity and differentially so across firms with and without US dollar debt.

5.2.1. US monetary policy transmission

In the first step of our analysis, we study how US monetary policy transmits abroad via financial channels other than the balance sheet

Table 2
US monetary spillover without interventions (Controlling for financial channels).

(a) Firms without US dollar debt

Dependent variable	ΔStock Price$_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	−0.094**	−0.030	−0.036	−0.093**	−0.094**	0.009
	(0.045)	(0.034)	(0.044)	(0.045)	(0.045)	(0.037)
R^2	0.032	0.044	0.042	0.035	0.032	0.054
N	103,155	103,155	95,668	103,155	103,155	95,668
VIX		✓				✓
Currency risk premium			✓			✓
Foreign yield				✓		✓
Capital flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

(b) Firms with US dollar debt

Dependent variable	ΔStock Price$_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	−0.660***	−0.626***	−0.580***	−0.666***	−0.549***	−0.552***
	(0.117)	(0.109)	(0.117)	(0.111)	(0.126)	(0.130)
R^2	0.093	0.104	0.113	0.094	0.100	0.127
N	1926	1926	1724	1926	1926	1724
VIX		✓				✓
Currency risk premium			✓			✓
Foreign yield				✓		✓
Capital flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), we control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We take the sample firms in countries without counter-interventions. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 3
Balance-sheet channel: Firms with dollar debt (Controlling for financial channels).

Dependent variable	ΔStock Price$_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	−0.647***	−0.561***	−0.600***	−0.652***	−0.558***	−0.487***
	(0.116)	(0.100)	(0.120)	(0.112)	(0.131)	(0.127)
FFR Shock$_t$ × Intervention$_{c,t}$	0.449***	0.386***	0.415***	0.452***	0.360**	0.330**
	(0.130)	(0.144)	(0.132)	(0.129)	(0.142)	(0.152)
R^2	0.091	0.100	0.111	0.092	0.097	0.120
N	3206	3206	2635	3206	3206	2635
VIX		✓				✓
Currency risk premium			✓			✓
Foreign yield				✓		✓
Capital flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), we control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We take the sample firms with dollar debt. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

channel. We re-estimate Eq. (2), where now the dependent variable is the VIX, the currency risk premium, foreign yields, and portfolio flows.

To begin with, US monetary policy shocks affect investors' risk appetite, measured by the VIX, and drive the "Global Financial Cycle" (Rey, 2015; Miranda-Agrippino and Rey, 2020). A contractionary monetary policy shock increases the investors' risk aversion, which in turn leads to global capital retrenchment and stock price declines. Following the literature (Bekaert et al., 2013; Bauer et al., 2023), we use the VIX as an indicator of risk aversion.

Furthermore, US monetary tightening not only increases investors' risk aversion in the stock market but also reduces investors' willingness to assume foreign currency risk. Since the dollar tends to appreciate

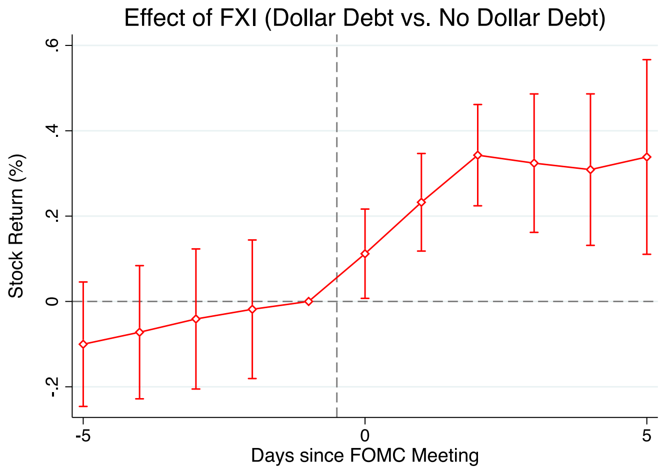

Fig. 3. Stock price: Triple interaction.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Eq. (5) for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

in "bad" times, foreign currency assets with lower returns become less attractive because they do not sufficiently hedge against increased marginal utility. In stochastic discount factor (SDF) terms, the negative covariance between the asset's return (here, the foreign currency return) and the SDF increases, leading investors to demand a higher excess return, i.e., a risk premium on foreign currency exposures. This elevated risk premium, in turn, lowers the stock price today relative to the future.[32][33]

Fig. 4, panel (b) shows that, without intervention, the currency risk premium increases in response to contractionary monetary policy shocks, but the currency risk premium does not increase when countries counter-intervene. In contrast to the VIX, we can leverage cross-country data to see whether the foreign currency risk premium varies across countries. In particular, FXI can mitigate the exchange rate risk and reduce the volatility of the foreign currency return. Hence, investors become more willing to invest in foreign currencies and require less compensation for taking foreign exchange risks. In SDF terms, FXI decreases the negative covariance between the foreign currency return and the SDF, making investors require a lower risk premium on foreign currency exposures.[34]

In addition, US monetary policy shocks may influence foreign stock prices via changes in the yield curve. Without interventions, a US monetary tightening shock depreciates the foreign exchange rate, raising the expected foreign short-term policy rate. This higher expected short-term interest rate can also raise the entire foreign yield curve—thereby raising the risk-free rate used for discounting foreign stock cash flows (Hanson and Stein, 2015; Nagel and Xu, 2024).[35] FXIs can stabilize the exchange rate and thus mitigate the international spillovers of US monetary policy via changes in foreign yield curves. To proxy for this foreign currency risk-free rate channel, we employ 10-year sovereign yields. Empirically, the foreign bond yield channel is relatively muted and not much affected by FXIs, as can be seen in Fig. 4, panel (c).

Lastly, contractionary US monetary policy may prompt withdrawals from foreign currency portfolios, depressing stock prices through a portfolio rebalancing channel. We test this channel using investor flows to the countries in our sample around FOMC announcements. Panel (d) shows strong investor outflows from foreign funds as a response to a contractionary US monetary policy. However, consistent with the idea that FXIs reduce currency risk, those outflows are mitigated when the foreign country intervenes against the Fed.

5.2.2. *Stock market transmission*

These results raise the question of whether US monetary policy not only affects firms' stock prices through a balance sheet channel, by preventing the reduction in the net worth, but also can work through a risk premium channel or a portfolio rebalancing channel. In the next step of the analysis, we test whether foreign stock prices are indeed affected through those channels and how they affect our coefficients. As can be seen from Table 1, panel (a), without interventions, even firms without US dollar debt are affected by the US monetary policy shock. The channels outlined above should indeed affect those firms, while firms with US dollar debt are additionally affected through the balance sheet channel. In the next step of the analysis, we test whether foreign stock prices are indeed affected through the risk premium and portfolio rebalancing channels and how they affect our coefficients.

First, we restrict the sample to countries without interventions and study stock market spillovers of US monetary policy via risk premium and portfolio rebalancing channels. To do so, we include the VIX, currency risk premium, foreign bond yield, and equity portfolio flows as controls δ_d^h in Eq. (2):

$$y_{i(c),t+1} - y_{i(c),t-1} = \beta \mathrm{FFR}_t + \mathbf{X}\delta_x + \mathbf{D}\delta_d + \alpha_{i(c)} + \epsilon_{i(c),t} \tag{6}$$

Hence, we estimate Eq. (6) for firms with and without US dollar debt separately and in countries without interventions. As shown in Table 2, panel (a) (column 1), for firms without US dollar debt, the contractionary US monetary policy shock affects those firms' stock prices negatively as long as we do not control for the other channels. However, once we control for the above channels, the coefficient on the US monetary policy shock becomes insignificant (columns 2–6). In particular, currency risk premium and the equity risk aversion channel explain the negative effect of the US monetary policy shocks on firms' stock prices, while the effects of bond yield and portfolio rebalancing channels are muted. Instead, panel (b) of the same table focuses on the firms with US dollar debt. For those firms, the coefficient on the FFR shock decreases slightly once we control for the other channels, but remains statistically significant, suggesting the balance sheet channel

[32] Classical evidence shows that a US monetary tightening leads to a persistent dollar appreciation and higher excess return on the US dollar (Eichenbaum and Evans, 1995). In contrast, modern evidence shows that an increase in the US interest rate forecasts an immediate appreciation and a subsequent depreciation of the dollar, implying higher excess return on the foreign currency (Engel, 2016). Literature explains this pattern using a model where the US supplies safe assets during crisis times (Maggiori, 2017; Hassan et al., 2023; Jiang et al., 2021, 2024).

[33] Note that this relationship only holds for currencies with a positive risk premium. In the following analysis, if we run a regression for the Japanese yen and Swiss franc, which have a negative currency risk premium, the results do not hold.

[34] In Fig. 4, we separate the sample into dates with and without interventions. In Fig. A.3, we instead take the interaction between the FFR shock and intervention and confirm that the effects of FXI on the risk premiums and capital flows are statistically significant.

[35] Hanson and Stein (2015) explain this channel using a model where investors' objective function depends not only on future expected returns but also on current income or yield. An increase in the short-term rate prompts investors to rebalance their portfolio from long-term to short-term bonds, generating upward pressure on long-term bond yields.

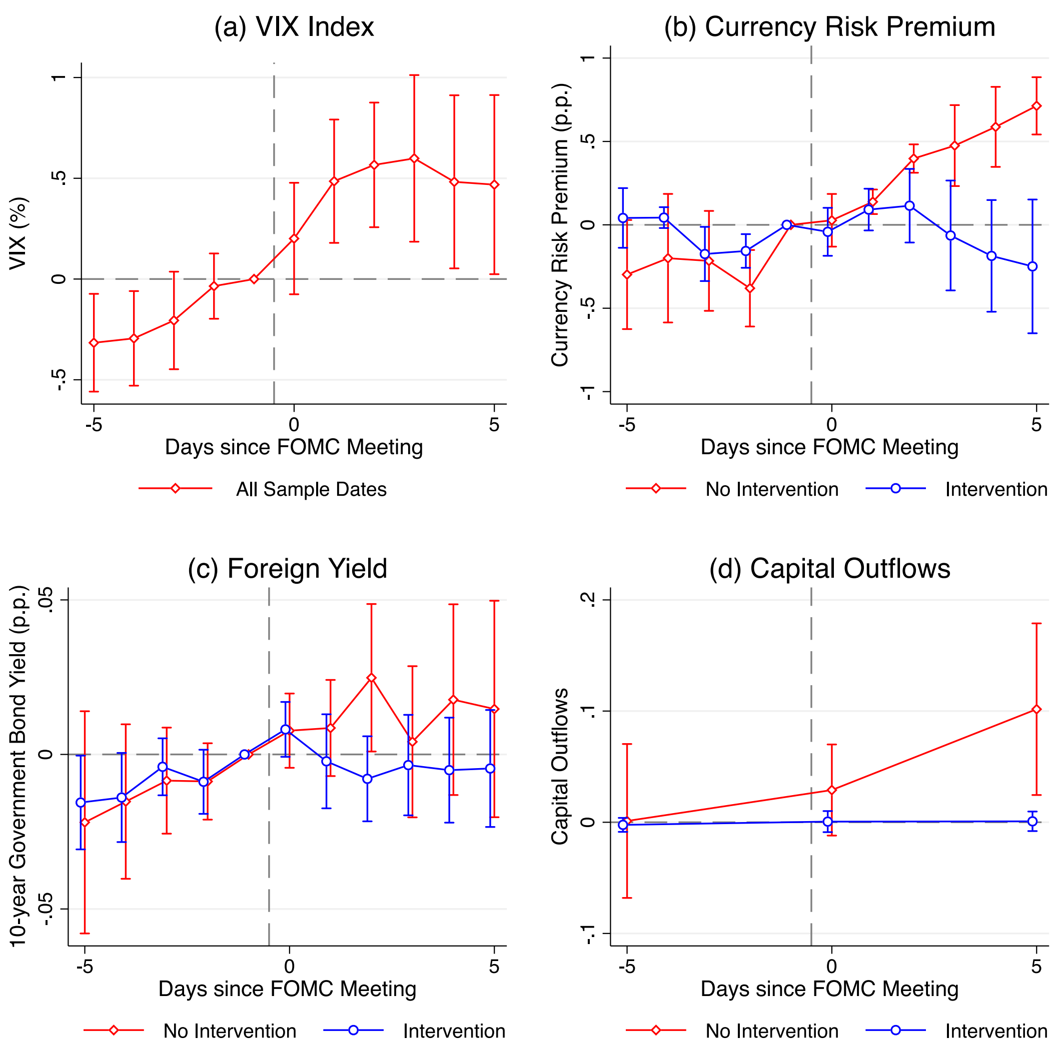

Fig. 4. Different transmission channels of FXI.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on the VIX index, currency risk premium, foreign bond yield, and equity capital flows. In panel (a), the dependent variable is the change in the 3-month US VIX index. In panel (b), the dependent variable is the change in currency risk premium. A positive value implies a higher excess return on the foreign currency over the US dollar. In panel (c), the dependent variable is the change in foreign 10-year government bond yields. In panel (d), the dependent variable is the equity capital flows. A positive value implies a capital outflow from foreign countries. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level (panel b) and clustered at the date level (panels a, c, and d). Note that, given low clusters due to data availability, we cannot double cluster in panels (c) and (d). The confidence interval is 90%.

plays an important role for those types of firms.[36] Next, we test whether those channels interact with the effectiveness of FXIs. We study whether the effect of FXIs on firms with dollar debt can solely be attributed to the balance sheet channel or also to the risk premium and portfolio rebalancing channels. To do so, we restrict the sample to firms with dollar debt and re-estimate Eq. (4), controlling for other channels:

$$y_{i(c),t+1} - y_{i(c),t-1} = \beta \mathrm{FFR}_t + \Omega FXI_{c,t} + \gamma \mathrm{FFR}_t \times FXI_{c,t} + \mathbf{X}\delta_x + \mathbf{D}\delta_d + \alpha_{i(c)} + \epsilon_{i(c),t} \tag{7}$$

As shown in Table 3, column (1), if we do not control for other channels, the FFR shock lowers the stock price of firms without dollar debt, and FXIs mitigate this stock price decline. Columns (2)–(6) show that, when we control for other channels, both the coefficients on FFR_t and $\mathrm{FFR}_t \times FXI_{c,t}$ become slightly smaller in magnitude. However, the estimates remain statistically significant, and we cannot reject the null hypothesis that the estimates in column (1) and columns (2)–(6) are about the same.

In sum, "intervening against the Fed" mitigates stock price declines not only via the balance sheet channel but also via the currency risk premium and portfolio rebalancing channels. However, even after controlling for the currency risk premium and portfolio rebalancing channels, the balance sheet channel of FXI remains intact.

5.3. Exchange rate

To further understand the balance sheet channel, we study the effect of FXIs on the exchange rate. Our previous results suggest that US

[36] This can also be seen in Table A.5, which shows the results for the whole sample and focuses on the interaction effect between the FFR shock and dollar debt. The baseline coefficient diminishes once we include those controls, while the coefficient on the interaction term remains statistically significant. This indicates that the "direct" effect of contractionary US monetary policy shock goes through the above channels, while the balance sheet channel remains statistically significant, explaining the differential effect.

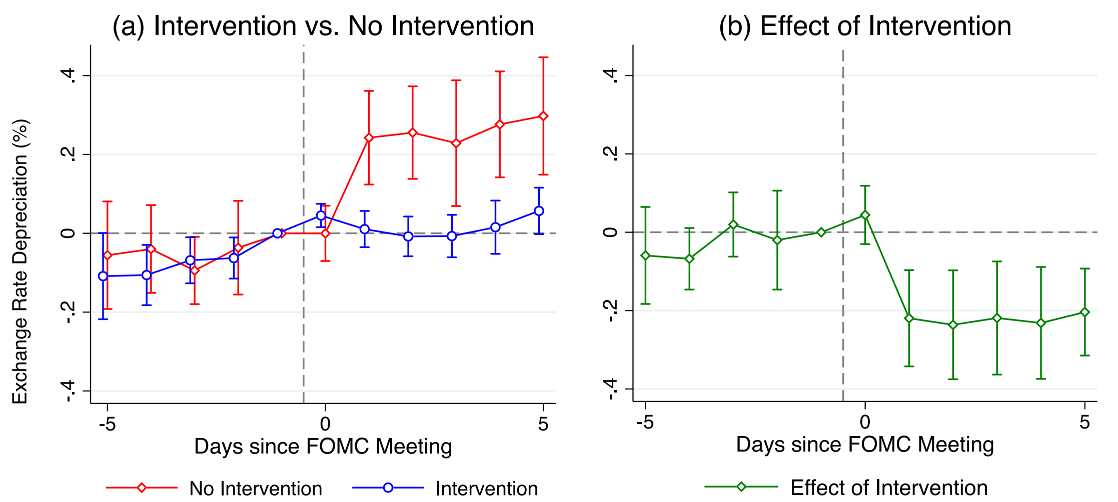

Fig. 5. Exchange rate: Difference-in-differences.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on exchange rates. See Eq. (4) for the exact specification. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level. The confidence interval is 90%.

monetary tightening without FXIs has a negative balance sheet effect on firms with dollar debt, but FXIs can mute this spillover. For this result to be true, it must be the case that a US monetary tightening depreciates local exchange rates when countries do not counter-intervene, but does not depreciate them when they counter-intervene. This is because the local depreciation increases the repayment of US dollar debt in terms of local currency and tightens the balance sheet of firms with US dollar debt.

To check this hypothesis, we estimate Eq. (2), where now the dependent variable is replaced with the change in the log of the exchange rate in the country c between dates $t-1$ and $t+1$. The exchange rate is defined as the value of one US dollar in terms of local currency, so that higher values imply an appreciation of the US dollar or a depreciation of local currency. $\beta > 0$ implies that a surprise tightening of US monetary policy depreciates the local exchange rate h days after the meeting. We control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, the one-month lagged policy rate, one-year lagged GDP, inflation, the trade balance over GDP ratio, the unemployment rate, and their interaction with the FFR shock.

In Table 4, columns (1) and (2) show the results for countries with and without intervention, respectively. If the local monetary authorities do not intervene, when the Fed unexpectedly hikes interest rates, the domestic currency depreciates in a statistically significant manner (by 2.25% in response to a 10 bp surprise hike). However, if local policymakers counter-intervene in the FX market by selling the US dollar, the exchange rate remains flat around the FOMC meetings. To check that the effect of FXIs is large enough, in column (3), we estimate Eq. (4) after replacing the dependent variable with a change in the exchange rate. Without interventions, a 10 bp surprise hike in the Fed funds rate leads to a 2.01% exchange rate depreciation. However, if central banks counter-intervene by selling the US dollar, the depreciation is reduced by 2.02% compared to a scenario without any counter-interventions, and this difference is statistically significant. Thus, interventions offset the exchange rate depreciation caused by the Fed's surprise.

To study how the effect of FXIs accumulates over time, we re-estimate Eq. (2) for all $h =\in [-5, 5]$. Fig. 5, panel (a) plots the estimates of β_h coefficient. Before the FOMC meeting, there is little difference in exchange rates between countries with and without FXIs. However, while countries that do not counter-intervene by selling the US dollar experience persistent depreciation, those that counter-intervene do not experience depreciation. The depreciation builds up slightly over time, consistent with Roussanov and Wang (2023). Moreover, to compare the trends of exchange rates around FOMC meetings in countries with and without FXIs, panel (b) re-estimates the γ_h coefficient in Eq. (4) for the exchange rate. $\gamma_h < 0$ for $h > 0$ implies that the exchange rate depreciates less when the central banks counter-intervene against

Table 4
Exchange rate: Baseline regression.

Dependent variable:	ΔExchange Rate$_{c,t}$		
	No intervention (1)	Intervention (2)	Both (3)
FFR Shock$_t$	0.225***	0.004	0.201**
	(0.069)	(0.021)	(0.072)
Intervention$_{c,t}$			0.266
			(0.155)
FFR Shock$_t$ × Intervention$_{c,t}$			−0.202**
			(0.072)
R^2	0.108	0.083	0.084
N	418	417	836
Country FE	✓	✓	✓

Note: ΔExchange Rate$_{c,t}$ is the change in the log of the exchange rate from date $t-1$ to $t+1$, where t is the FOMC announcement date. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. We control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, the one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the country and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

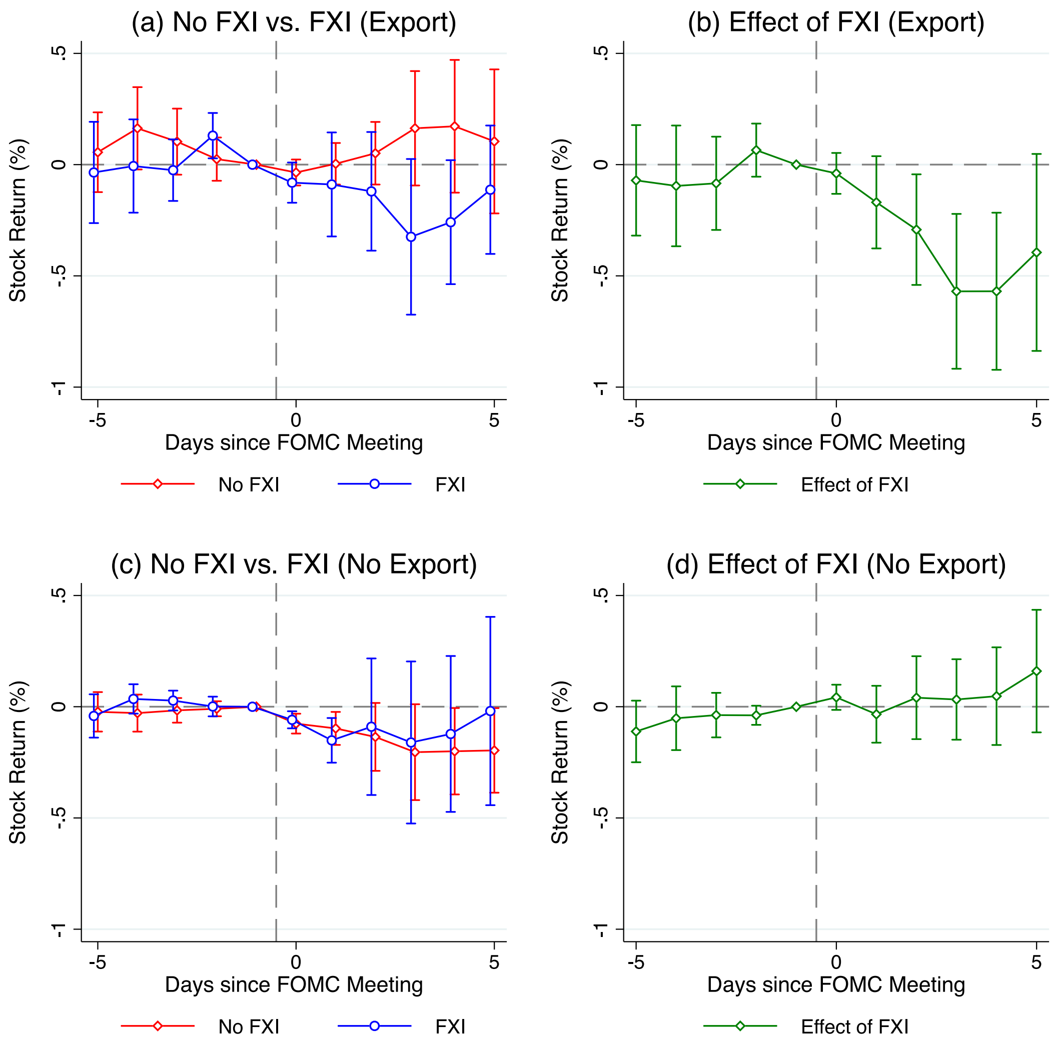

Fig. 6. Stock price: Expenditure switching channel.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices for exporters and non-exporters. See Eqs. (3) and (5) for the exact specifications. We control for one-year lagged dollar debt, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

the Fed compared to a no-intervention case.[37] These results suggest that FXIs are successful in stabilizing the exchange rate in response to unexpected monetary shocks, giving further support to the balance sheet stabilization channel of FXIs for firms with dollar debt.

5.4. Expenditure switching channel

In the previous sections, we have shown that FXIs benefit firms with dollar debt by muting the negative balance sheet channel of US monetary tightening. However, a US tightening may also have other spillover effects, most importantly via exporting firms. In this section, we first study the effect of US monetary tightening and FXIs on exporters and non-exporters. Next, we test whether exports, as well as international sales and assets, can explain the negative effects of FXIs on firms without dollar debt.

US monetary tightening has two offsetting effects on exporting firms. On the one hand, depreciation induced by contractionary US monetary policy may increase foreigners' demand for exports due to lower relative prices (expenditure switching channel). On the other hand, a US monetary tightening may reduce US demand for goods and thus the demand for exports (a negative demand channel). If FXIs mute the depreciation-induced expenditure switching channel but do not mute the negative demand channel due to intertemporal substitution, we would expect FXIs to be costly for exporters. In addition, since almost all exporting firms in the sample do not have dollar debt (Appendix A), the balance sheet channel is likely muted, and the expenditure switching channel dominates.

To check this possibility, we study the effect of FXIs on exporters and non-exporters. In Fig. 6, we conduct a similar exercise to Fig. 2 for exporters and non-exporters. Interestingly, although we do find some suggestive evidence for expenditure switching channels, the effect is quantitatively small. Fig. 6, panel (a) studies the stock price response to Fed hikes for exporters in countries with and without FXIs, respectively. When the Fed tightens, the stock price of exporters increases without FXIs but decreases with FXIs. However, the magnitude of the stock price decline is small (3.2% at the trough in response to the 10 bp US hike) and the statistical significance is low. In contrast, Fig. 2 suggests that stock prices for firms with dollar debt decrease significantly

[37] The effect of FXIs is statistically significant one day after the interventions. This is potentially due to the difference in time zones. Since the United States is one of the most Western countries in the world, the FOMC announcement does not affect the exchange rate in Eastern countries, such as Japan, on the same date. The limitation of our research is that we do not have intra-day data on the exact timing of FXIs in each country.

Table 5
Stock price (Excluding firms with foreign revenues and assets).

(a) Without intervention

Dependent variable	ΔStock Price$_{i(c),t}$				
	(1)	(2)	(3)	(4)	(5)
FFR Shock$_t$	−0.097**	−0.095**	−0.089**	−0.091*	−0.082*
	(0.045)	(0.044)	(0.044)	(0.046)	(0.043)
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$	−0.314***	−0.316***	−0.292***	−0.313***	−0.313***
	(0.087)	(0.088)	(0.082)	(0.088)	(0.085)
R^2	0.031	0.032	0.035	0.035	0.036
N	105,114	96,056	93,608	97,478	83,128
Exclude firms with exports		✓			✓
Exclude firms with international sales			✓		✓
Exclude firms with international assets				✓	✓
Firm FE	✓	✓	✓	✓	✓

(b) With intervention

Dependent variable	ΔStock Price$_{i(c),t}$				
	(1)	(2)	(3)	(4)	(5)
FFR Shock$_t$	−0.158**	−0.157**	−0.089	−0.072	−0.085
	(0.061)	(0.062)	(0.082)	(0.049)	(0.077)
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$	−0.001	0.002	−0.044	−0.015	−0.041
	(0.042)	(0.045)	(0.081)	(0.059)	(0.086)
R^2	0.194	0.186	0.257	0.205	0.246
N	11,178	10,640	6153	9713	5606
Exclude firms with exports		✓			✓
Exclude firms with international sales			✓		✓
Exclude Firms with International Assets				✓	✓
Firm FE	✓	✓	✓	✓	✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. In column (2), we exclude firms with exports. In column (3), we exclude firms with international sales. In column (4), we exclude firms with international assets. In column (5), we exclude firms with exports, international sales, and international assets. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

without FXIs (9.0% at the trough), and FXIs mitigate this decline. Next, panel (b) plots the interaction coefficient between FXIs and the export indicator. We show that FXIs decrease the stock price response for exporters compared to cases without FXIs. However, the effect is statistically significant only two days after the FOMC announcement, and the significance disappears five days after the event. In contrast, Fig. 2, panel (b) suggests that FXIs have an immediate positive impact on firms with dollar debt on the FOMC announcement date, and the effect is persistent over time. Our estimates suggest that, although FXIs have the cost of mitigating expenditure switching channels for firms that export, the benefits for firms with US dollar debt are larger. Ultimately, it may depend on the composition of firms whether FXIs increase or decrease stock prices in the aggregate. For countries in which a large share of firms borrow in US dollars, the positive effect may dominate, but for countries with a large exporting sector, the negative effects could be substantial.

Next, we study what explains the negative effects of FXIs on firms without dollar debt, as shown in Table 1 of Section 5.1. One possibility is the expenditure switching channel for exporting firms. As discussed in Appendix A, most of the exporting firms in our sample do not have dollar debt, so the balance sheet channel is muted for exporting firms. Hence, "Intervening against the Fed" may negatively affect the stock price of firms with exports but without dollar debt. In addition to exports, US monetary policy and FXIs may affect firms with international sales or assets.[38] US monetary tightening may benefit these firms by appreciating the US dollar and increasing the value of international sales or assets in terms of the local currency. However, FXIs may harm these firms by appreciating the local currency and mitigating this positive valuation effect.

In Table 5, we test which channels described above can explain the negative effects of FXIs on firms without US dollar debt. Panels (a) and (b) show the results without FXI and with FXI, respectively. If we do not exclude firms with exports, international sales, and international assets (column 1), the coefficient on the FFR_t term is larger with intervention ($\beta = -0.158$, panel b) than without intervention ($\beta = -0.097$, panel a). In columns (2)–(4), we exclude firms with exports, international sales, and international assets, respectively. In column (5), we exclude all of these firms. The table shows that if we exclude firms with international sales or assets, the negative coefficient on the FFR_t term with interventions becomes statistically insignificant (columns 3–5 of panel b) and no longer different from the case without interventions (columns 3–5 of panel a) in a statistically significant manner. This result suggests that the negative effect of interventions on firms without dollar debt can be explained by firms with international sales or assets.

Our results suggest an important policy tradeoff. Intervening against the Fed stabilizes currency risk premia, portfolio flows, and the stock prices of firms, especially those with US dollar debt. However, interventions can also harm firms with international revenues or assets. Whether central banks should counter-intervene or not depends on the relative strengths of these different channels.

We acknowledge that we do not find evidence that FXI benefits firms without dollar debt, which constitute the majority of our sample firms.[39] As discussed in Section 3.1, the limitation of our analysis is

[38] Exports are defined as sales of domestically produced goods in foreign countries, while international sales are defined as sales of goods in foreign countries, regardless of whether goods are produced domestically or abroad.

[39] We also do not find evidence that FXI affects the country-level stock market index, which measures the overall stock market performance.

Table 6
Stock price: FXI volume.

Dependent variable	ΔStock Price$_{i(c),t}$	
	(1)	(2)
FFR Shock$_t$	−0.097**	
	(0.045)	
FFR Shock$_t$ × Intervention$_{c,t}$	0.005	
	(0.063)	
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$	−0.152***	−0.167***
	(0.044)	(0.046)
FFR Shock$_t$ × Intervention$_{c,t}$ × Dollar Debt$_{i(c),y-1(t)}$	0.142**	0.111**
	(0.058)	(0.051)
R^2	0.034	0.086
N	114,448	114,448
Firm FE	✓	✓
Country × Date FE		✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to t+1, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is the standardized volume of unexpected intervention. The FXI volume is measured in terms of the percentage ratio over GDP. A positive value indicates a net purchase of the local currency. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

that it is difficult to obtain firm-level data on the invoicing currency and destination of exports, international sales, and international assets for a panel of multiple countries. Another limitation is that Worldscope has missing values for exports. In this paper, we focus on the balance sheet heterogeneity since we have detailed firm-level data on the currency composition of balance sheets. We believe that improved data availability in the future would allow for more accurate testing of FXI transmission channels across dimensions of firm heterogeneity.

5.5. Size of intervention

While we mainly use the $(0, 1, -1)$ categorical variable for FXIs, policymakers may be interested in how much volume of FXIs is sufficient to offset the effect of US monetary policy shocks on the stock market. To address this question, we replace this categorical variable with the volume data of FXI. The FXI volume is measured as a percentage ratio over GDP, and a positive value implies a net purchase of the local currency (net sales of the US dollar). We standardize the FXI volume so that the mean is zero and the standard deviation is unity. We have replaced the categorical variable with the intervention volume data, both in the first- and second-stage regressions.

Table 6 shows that our result is robust to using volume data of interventions instead of a categorical variable. Column (1) shows that a 10 bp increase in the Fed funds rate lowers the stock price of firms with US dollar debt by 1.5 percent, compared to firms without US dollar debt. However, one-standard-deviation US dollar sales (90 million dollars) mitigate this stock price decline by 1.4 percent.[40] In other words, US dollar sales of 101 million dollars fully offset the stock market transmission of US monetary policy shocks via the balance sheet channel. This result is robust after controlling for the country-time fixed effect (column 2).

We compare this estimate with the actual FXI volumes by central banks. We take the standard deviation of FXI volumes across all unexpected counter-intervention episodes against US monetary shocks. Across all our sample countries, one standard deviation of FXI equals 139 million dollars. If we focus on each country, one standard deviation of FXI equals 114 million dollars in Argentina and 124 million dollars in Brazil. Hence, our estimate is close to the actual volume of counter-interventions by central banks, especially in emerging economies. These results suggest that the volume of FXI is quantitatively meaningful in stabilizing the stock market spillovers of US monetary shocks via balance sheets.

The reason we use a $(0, 1, -1)$ categorical variable throughout the paper is to mitigate the endogeneity. As discussed in Ito and Yabu (2007) and Menkhoff et al. (2021), the decision on whether to intervene or not depends on general financial market and economic conditions in the medium to long run. In contrast, the decision on how much to intervene depends on contemporaneous information on exchange rates or asset prices. We follow the literature and use the categorical variable as a baseline measure to minimize this endogeneity concern.

5.6. Foreign exchange reserves

Central banks' FX reserves have increased significantly over the last decades. Fig. A.4 shows the volume of FX reserves and the FX reserves over GDP ratio in our sample of countries. In the 1990s, FX reserves were low (roughly 0.21 trillion US dollars), and countries had limited instruments to insulate themselves from US monetary spillovers. However, reserves have increased by a magnitude of around 18 since the 1990s, reaching a peak of 3.91 trillion dollars in 2021. The FX reserve-to-GDP ratio also grew from 4.3% in 1990 to a peak of 32.5% in 2020. So far in 2022, the tightening of US monetary policy has not yet had severe negative consequences for emerging market economies, raising the question of whether FXIs are more effective when countries have relatively large FX reserves, as they can sell off those accumulated reserves to prevent sudden depreciations of exchange rates during times of crisis.

To test this hypothesis, we study the effects of FXIs on exchange rates and stock prices in countries with heterogeneous degrees of FX reserves. We use annual data on total FX reserves in the IMF International Financial Statistics.[41] We define large and small FX reserves if the volume of FX reserves is larger or smaller than the median, respectively.

In Table 7, columns (1) and (2) estimate Eq. (4) for exchange rates in countries with large and small reserves, respectively. We find that, in countries with large FX reserves, a 10 bp Fed hike leads to a 2.5% depreciation in the exchange rate, but the depreciation is fully stable when the country intervenes. However, in countries with small FX reserves, the effect of FXIs is reduced to 1.6% and it is statistically less significant. Similarly, columns (3) and (4) estimate Eq. (5) for stock prices of firms based in countries with large and small reserves, respectively. An increase in the Fed funds rate leads to a decline in stock prices for firms with dollar debt relative to those without dollar debt. FXIs mitigate this balance-sheet spillover in countries with large FX reserves (4.2pp), but the effect is weaker in countries with small FX reserves (1.3pp). Columns (5) and (6) show this result is robust even after including country-date fixed effects. These results suggest that FX reserve accumulation plays a crucial self-insurance role in mitigating US monetary spillovers.

[40] The intervention of 90 million dollars corresponds to around 0.03% of GDP when evaluated at the median level of GDP across all samples.

[41] The dataset provides total FX reserves at a country level, but the currency composition of reserves is only available at the world level. Iancu et al. (2022) and Ito and McCauley (2020) provide data on the currency composition of FX reserves. Since the currency composition data is not available for all of our sample countries, we use the total FX reserves data.

Table 7
FX reserves.

Dependent variable:	ΔExchange Rate$_{c,t}$		ΔStock Price$_{i(c),t}$			
FX reserve:	Large (1)	Small (2)	Large (3)	Small (4)	Large (5)	Small (6)
FFR Shock$_t$	0.252***	0.122	−0.075	−0.126		
	(0.065)	(0.071)	(0.067)	(0.094)		
FFR Shock$_t$ × Intervention$_{c,t}$	−0.298***	−0.161*	−0.108	0.162		
	(0.089)	(0.080)	(0.079)	(0.143)		
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$			−0.387**	−0.261***	−0.352***	−0.230***
			(0.151)	(0.086)	(0.108)	(0.079)
FFR Shock$_t$ × Intervention$_{c,t}$ × Dollar Debt$_{i(c),y-1(t)}$			0.418**	0.137	0.317***	0.201*
			(0.172)	(0.112)	(0.118)	(0.102)
R^2	0.121	0.149	0.053	0.051	0.109	0.098
N	422	413	90,860	25,880	90,860	25,880
Country FE	✓	✓				
Firm FE			✓	✓	✓	✓
Country × Date FE					✓	✓

Note: Columns (1) and (2) show the results for country-level exchange rates, and columns (3) to (6) show the results for firm-level stock prices. ΔExchange Rate$_{c,t}$ is the change in the log of the exchange rate from date $t-1$ to $t+1$, and ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. Large and small FX reserves are defined so that the volume of FX reserves is larger and smaller than the median, respectively. In columns (1) and (2), we control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. In columns (3)–(6), we control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

5.7. Robustness checks

Appendix A provides various robustness checks to confirm our result, including debt maturity, intensive and extensive margins of dollar debt, alternative specification for unexpected counter-interventions, sterilized interventions, including periods with zero FFR shock, excluding global financial crisis and zero lower bound on the interest rate, international sales and asset holdings, currency denomination of stock price, and currency risk hedging. Furthermore, we control for heterogeneous cross-country characteristics that can confound our estimations, including advanced and emerging economies, capital account openness, trade openness, and development. Overall, the key economic implication of the balance sheet channel of FXIs remains robust under these different settings.

6. Conclusion

US monetary policy has significant spillover effects on other countries. In this paper, we investigate those mechanisms of US monetary policy using a high-frequency approach that allows us to understand the effects and channels through which monetary policy affects foreign economies more directly. When countries do not intervene, an unexpected US monetary policy tightening induces foreign exchange rate depreciation against the US dollar, a rise in the currency risk premium, portfolio outflows, and declines in firms' stock prices. The stock prices decline significantly more for firms that borrow in US dollars, mirroring the experience of earlier episodes, such as the 1990s or during the taper tantrum.

In the 1990s, central bank holdings of foreign exchange reserves were low, and they had limited ability to protect themselves against the spillovers of US monetary policy. However, since the 1990s to today, FX reserves have grown by a magnitude of around 16, as shown in Fig. A.4, likely in anticipation that those reserves can act as a self-insurance mechanism.

In this paper, we have shown that "intervening against the Fed" mitigates exchange rate depreciation, the rise in risk premiums, and portfolio rebalancing. When US monetary policy tightens and central banks counter-intervene, changes in exchange rates, currency risk premiums, and portfolio flows remain stable. Furthermore, interventions mitigate stock price declines for firms, with disproportionately stronger effects for firms with US dollar debt. These results suggest that intervening against the Fed mutes the balance sheet channel of exchange rates triggered by US monetary policy and can protect countries from exposure to the Global Financial Cycle.

Overall, countries today should be less vulnerable to US monetary policy than in previous tightening cycles. However, some countries' reserves remain low, especially those of low-income countries, and their ability to protect themselves from the global financial cycle remains limited.

While we do not study the optimality of the policy interventions, in theory, intervening against the Fed can be optimal if US monetary policy triggers a risk appetite shock (Basu et al., 2020), pecuniary externalities (Fanelli and Straub, 2021), or local inflationary pressures (Yago, 2024). Further understanding general equilibrium implications and the optimality of policies is an important agenda for future research.

CRediT authorship contribution statement

Alexander Rodnyansky: Conceptualization. **Yannick Timmer:** Writing – review & editing, Writing – original draft, Conceptualization. **Naoki Yago:** Formal analysis.

Declaration of competing interest

The authors has nothing to disclose.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2026.104265.

Data availability

Replication Package for "Intervening against the Fed" (Original data) (Mendeley Data)

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